

Technical Report: Distributed Asynchronous Pose Graph Optimization over Planar Unit Dual Quaternions

Abstract—Distributed pose graph optimization helps enable multi-robot localization. However, many existing approaches assume that uncertainties in robots’ positions and orientations are decoupled, which often does not hold in practice. Therefore, in this work we develop a distributed algorithm for solving pose graph optimization problems with coupled, anisotropic uncertainties. We study two-dimensional pose graph optimization, and we model planar robot poses on the manifold of planar unit dual quaternions. Then we develop a new distributed Riemannian Levenberg-Marquardt method on this manifold in which robots’ computations and communications are asynchronous, subject only to bounded communication delays. We prove that this algorithm converges to first-order critical points of the robots’ objective from all initializations. Experiments demonstrate that, under anisotropic uncertainties, our algorithm reduces pose graph estimation error by 3.6% to 20.4% compared to state-of-the-art algorithms.

I. INTRODUCTION

Multi-robot systems are applied in complex environments for tasks ranging from search and rescue [1], [2] to cooperative exploration [3], [4]. In unknown environments, robots often use collaborative simultaneous localization and mapping (CSLAM) [5]–[9] to build a shared map via pose graph optimization (PGO). While centralized PGO solvers such as SE-Sync [10], Cartan-Sync [11], and CPL-Sync [12] are well established, they suffer from single points of failure and communication bottlenecks, which are undesirable for multi-robot systems. Consequently, there is a need for decentralized architectures that do not rely on a central coordinator.

To address these challenges, decentralized PGO algorithms have been developed. Synchronous methods [13] force robots to wait for the slowest among them, which can delay convergence. Asynchronous solvers like AS-APP [14] and the framework in [15] resolve this bottleneck but assume isotropic uncertainty, which limits accuracy in applications with anisotropic uncertainty [16], such as rotation averaging [17] and pose estimation [18], [19]. Although methods like MESA [20] and DVM-SLAM [21] allow anisotropic noise, they lack proofs of convergence. Majorization-minimization (MM) frameworks have been proposed to provide rigorous convergence guarantees for distributed PGO [22], but only for simplified PGO approximations rather than the exact PGO problem.

In this paper, we formulate the distributed PGO problem on the manifold of planar unit dual quaternions (PUDQs), and we use PUDQs to develop an anisotropic uncertainty model. Then, we solve the exact PGO problem without approximating it. We develop a distributed Riemannian Levenberg-Marquardt (RLM) algorithm that employs a damping parameter adapted to the manifold’s local curvature,

which helps to ensure global convergence to first-order critical points under asynchronous computations and communications. We consider planar (2D) pose graph optimization problems in which each robot pose lies in $SE(2)$, which we represent using planar unit dual quaternions.

The PUDQ formulation allows for coupled anisotropic translational-rotational uncertainty, which allows us to model pose uncertainty more accurately than with decoupled, isotropic noise models. Moreover, certain operations used by Riemannian optimization techniques (such as mapping off-manifold points to a manifold) have been shown to be more numerically efficient for PUDQs than other representations of $SE(2)$ [23], which further motivates the use of PUDQs.

Related work in [24] formulates planar PGO on the manifold of PUDQs and develops a provably convergent *centralized* Riemannian optimization algorithm. The present work fundamentally differs because it develops a distributed asynchronous algorithm for solving PGO problems over the PUDQs. Therefore, while we use some results from [24] to analyze our objective function, the centralized convergence analysis from [24] does not apply to the distributed asynchronous setting in this work.

To the best of our knowledge, this paper presents the first distributed asynchronous PGO algorithm that provably converges with anisotropic uncertainty, and our approach achieves global convergence from any initialization and with any level of measurement noise.

A. Summary of Contributions

The main contributions of this paper are:

- 1) We develop the first distributed asynchronous pose graph optimization algorithm formulated on the manifold of planar unit dual quaternions that explicitly accounts for anisotropic uncertainties (Algorithm 1).
- 2) We prove that, under bounded communication delays, the proposed algorithm achieves global convergence to first-order critical points from any initialization, and we derive a convergence rate (Theorems 1 and 2).
- 3) We simulate our approach on standard benchmarks, and we show a 3.6% to 20.4% reduction in error relative to both centralized PGO algorithms and distributed asynchronous PGO algorithms (Section VI).

The rest of the paper is organized as follows. Section II gives background, and Section III formulates the multi-robot PGO problem. Section IV develops the distributed RLM algorithm, and Section V proves its convergence. Then Section VI presents simulations, and Section VII concludes.

Notation We use $\mathbb{N}$ to denote the positive integers and $\mathbb{R}$ to denote the reals. Given a collection of vectors $v_1, v_2, \dots, v_N$, we write $(v_i)_{i=1}^N$ to denote the tuple $(v_1, v_2, \dots, v_N)$. For a tuple of vectors $(v_i)_{i=1}^N$, we define $\text{vec}((v_i)_{i=1}^N) = [v_1^\top, v_2^\top, \dots, v_N^\top]^\top$. For $n \in \mathbb{N}$ we define $[n] = \{1, 2, \dots, n\}$. We define $\text{sinc}(x) = \frac{\sin x}{x}$, where $\text{sinc}(0) = 1$. We write $\arctan$ for the four-quadrant arctangent function, and $\|\cdot\|$ denotes the Euclidean norm.

II. PRELIMINARIES

This section outlines the background for our approach.

A. PUDQ Representation of Rigid Motion

We use planar unit dual quaternions (PUDQs) to represent the group of planar rigid motions $\text{SE}(2)$ [25]. Given an orthonormal basis $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$, a planar rigid motion can be associated with a translation $\mathbf{t} = t_x \mathbf{i} + t_y \mathbf{j}$ for some $t_x, t_y \in \mathbb{R}$ and a rotation by $\theta \in (-\pi, \pi]$ about the $\mathbf{k}$ -axis. Such a motion can be encoded as a PUDQ $\mathbf{x} = \mathbf{x}_r + \varepsilon \mathbf{x}_d$, where ε is the dual unit satisfying $\varepsilon \neq 0$ and $\varepsilon^2 = 0$. The real component $\mathbf{x}_r \in \mathbb{S}^1$ can be computed with $\mathbf{x}_r \triangleq \cos(\theta/2) + \sin(\theta/2)\mathbf{k}$, and the dual component $\mathbf{x}_d \in \mathbb{R}^2$ is given by $\mathbf{x}_d \triangleq \frac{1}{2}\mathbf{t} \otimes \mathbf{x}_r$, where $\otimes$ denotes the standard Hamilton product of quaternions under the convention $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{i}\mathbf{j}\mathbf{k} = -1$. We sometimes call PUDQs “poses”.

For a PUDQ $\mathbf{x}$, we use the standard real-vector embedding $\mathbf{x} \mapsto [x_0, x_1, x_2, x_3]^\top \in \mathbb{R}^4$ so that the set of all PUDQs forms the smooth embedded manifold

$$\mathcal{M} \triangleq \{\mathbf{x} \in \mathbb{R}^4 \mid \mathbf{x}^\top \tilde{P} \mathbf{x} - 1 = 0\} \subset \mathbb{R}^4, \quad (1)$$

where $\tilde{P} \triangleq \text{diag}(1, 1, 0, 0)$. We embed $\mathcal{M}$ in $\mathbb{R}^4$, and therefore $\mathcal{M}$ inherits the metric $\langle v, w \rangle_{\mathbf{x}} = u^\top w$ and the norm $\|u\|_{\mathbf{x}} = \|u\|_2$.

The composition of two PUDQs $\mathbf{x}$ and $\mathbf{y}$ is denoted by $\mathbf{x} \boxplus \mathbf{y}$, and it takes the form

$$\mathbf{x} \boxplus \mathbf{y} = \underbrace{\begin{bmatrix} x_0 & -x_1 & 0 & 0 \\ x_1 & x_0 & 0 & 0 \\ x_2 & x_3 & x_0 & -x_1 \\ x_3 & -x_2 & x_1 & x_0 \end{bmatrix}}_{Q_L(\mathbf{x})} \underbrace{\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} y_0 & -y_1 & 0 & 0 \\ y_1 & y_0 & 0 & 0 \\ y_2 & -y_3 & y_0 & y_1 \\ y_3 & y_2 & -y_1 & y_0 \end{bmatrix}}_{Q_R(\mathbf{y})} \underbrace{\begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\mathbf{x}},$$

where $Q_L(\cdot)$ and $Q_R(\cdot)$ denote the left and right composition maps, respectively. The identity PUDQ is $\mathbf{1} = [1, 0, 0, 0]^\top$, and the inverse of a PUDQ $\mathbf{x} = [x_0, x_1, x_2, x_3]^\top$ is given by $\mathbf{x}^{-1} = [x_0, -x_1, -x_2, -x_3]^\top$.

To optimize over $N \in \mathbb{N}$ poses, we define the product manifold $\mathcal{M}^N$ and the trajectory $X = \text{vec}((\mathbf{x}_i)_{i=1}^N)$, where $\mathbf{x}_i \in \mathcal{M}$ for all $i \in [N]$. The metric and norm on the product manifold are also inherited from the ambient Euclidean space; see Appendix A for details. At X , the tangent space $\mathcal{T}_X \mathcal{M}^N$ is defined as the direct sum of individual tangent spaces, i.e.,

$$\mathcal{T}_X \mathcal{M}^N = \mathcal{T}_{\mathbf{x}_1} \mathcal{M} \oplus \mathcal{T}_{\mathbf{x}_2} \mathcal{M} \oplus \dots \oplus \mathcal{T}_{\mathbf{x}_N} \mathcal{M} \subset \mathbb{R}^{4N}. \quad (2)$$

B. Lie Exponential and Logarithm

The manifold $\mathcal{M}$ with the aforementioned group structure forms a Lie group with Lie algebra $\mathcal{T}_1 \mathcal{M}$, which is the tangent space to $\mathcal{M}$ at the identity $\mathbf{1}$. The Lie exponential maps from $\mathcal{T}_1 \mathcal{M}$ to $\mathcal{M}$, and its inverse is the Lie logarithm map¹. For a tangent vector $\tau = [\tau_1, \tau_2, \tau_3]^\top \in \mathcal{T}_1 \mathcal{M}$, the Lie exponential map $\exp : \mathcal{T}_1 \mathcal{M} \rightarrow \mathcal{M}$ takes the form

$$\exp(\tau) = [\cos(\tau_1), \text{sinc}(\tau_1) \tau^\top]^\top. \quad (3)$$

For a pose $\mathbf{x} \in \mathcal{M}$, the Lie logarithm map is $\log : \mathcal{M} \rightarrow \mathcal{T}_1 \mathcal{M}$, and it outputs the tangent vector

$$\log(\mathbf{x}) = \frac{1}{\text{sinc}(\phi(\mathbf{x}))} [x_1, x_2, x_3]^\top, \quad (4)$$

where $\phi(\mathbf{x}) \triangleq \text{wrap}(\arctan(x_1, x_0))$ and

$$\text{wrap}(\alpha) \triangleq \begin{cases} \alpha + \pi & \text{if } \alpha \leq -\pi/2 \\ \alpha - \pi & \text{if } \alpha > \pi/2 \\ \alpha & \text{otherwise} \end{cases}.$$

Leveraging these Lie-theoretic maps, we formulate an anisotropic noise model by mapping a zero-mean Gaussian distribution from the tangent space $\mathcal{T}_1 \mathcal{M}$ to $\mathcal{M}$ via the Lie exponential in (3).

C. Distributed Pose Graph Construction

We consider pose graphs that connect $R \in \mathbb{N}$ robots. We define $\mathcal{R} \triangleq [R]$, and we suppose that robot p has a trajectory of $N_p \in \mathbb{N}$ poses for each $p \in \mathcal{R}$. We model multi-robot pose graphs as directed graphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. The vertex set $\mathcal{V}$ is

$$\mathcal{V} = \bigcup_{p=1}^R \{(p, i) \mid 1 \leq i \leq N_p\},$$

where the vertex (p, i) corresponds to the i^{th} pose of robot p . The edge set $\mathcal{E}$ represents relative measurements between poses. An edge $((p, i), (p, j))$ is an *intra-robot edge* if it connects two poses of the same robot p , denoted $\mathbf{x}_i^p, \mathbf{x}_j^p \in \mathcal{M}$. These are called “unshared” poses because in Section IV robot p does not need to communicate its estimates of these poses with other robots. We write the edge $((p, i), (p, j))$ as (i, j) when p is clear from context, and $E_p^{\text{un}} \subseteq \mathcal{E}$ denotes the set of intra-robot edges of robot p .

For distinct $p, q \in \mathcal{R}$, an edge is an *inter-robot edge* if it connects poses $\mathbf{x}_i^p$ and $\mathbf{x}_j^q$ in either direction, i.e., both $((p, i), (q, j))$ and $((q, j), (p, i))$ can be inter-robot edges. Such poses are called “shared” because in Section IV robots p and q must communicate their estimates of these poses. We denote the set of inter-robot edges between robots p and q by $E_{pq}^{\text{sh}} \subseteq \mathcal{E}$, and this set contains all edges that connect robots p and q , regardless of their direction.

¹The Lie exponential and logarithm are distinct from the Riemannian exponential and logarithm, which are introduced in Section IV-A and Appendix A, respectively. We use the Lie-theoretic maps for defining probability distributions on the manifold’s group structure, and we use the Riemannian maps in our optimization algorithm.

We define the neighborhood set of robot p as

$$S(p) \triangleq \{q \in \mathcal{R} \setminus \{p\} \mid E_{pq}^{\text{sh}} \neq \emptyset\}, \quad (5)$$

which is the set of indices of other robots with which robot p shares an inter-robot edge in the pose graph $\mathcal{G}$. We define

$$\Delta \triangleq \max_{q \in \mathcal{R}} |S(q)|, \quad (6)$$

which is the greatest number of neighbors any robot has. Figure 1 shows an example of a multi-robot pose graph.

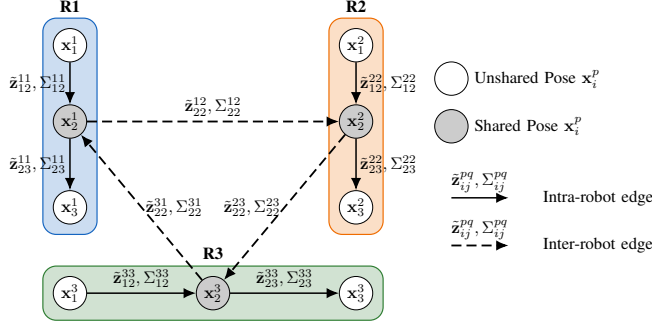


Fig. 1. Example of a three-robot pose graph $\mathcal{G}$. Each term $\tilde{z}_{ij}^{pq}$ represents a measurement of the relative transformation between poses $\mathbf{x}_i^p$ and $\mathbf{x}_j^q$, and Σ_{ij}^{pq} is its covariance matrix.

III. PROBLEM FORMULATION

We now formulate the multi-robot PGO problem we solve.

A. Measurement Model and Residuals

We adopt the Lie-theoretic measurement model from [24]. A relative measurement $\tilde{z}_{ij}^{pq}$ from pose $\mathbf{x}_i^p \in \mathcal{M}$ to pose $\mathbf{x}_j^q \in \mathcal{M}$ is modeled as the true relative pose corrupted by Gaussian noise η_{ij}^{pq} in the appropriate tangent space. Formally,

$$\tilde{z}_{ij}^{pq} = (\mathbf{x}_i^p)^{-1} \boxplus \mathbf{x}_j^q \boxplus \exp(\eta_{ij}^{pq}), \quad \eta_{ij}^{pq} \sim \mathcal{N}(0, \Sigma_{ij}^{pq}), \quad (7)$$

where $\exp$ is from (3). The matrix $\Sigma_{ij}^{pq} \in \mathbb{R}^{3 \times 3}$ is symmetric and positive definite, and it can be non-diagonal, which models anisotropic translational-rotational correlations in pose uncertainty. This formulation accurately represents the “banana-shaped” uncertainty characteristic of pose estimation as a Gaussian in exponential coordinates [26], [27].

Given a measurement $\tilde{z}_{ij}^{pq}$ between poses $\mathbf{x}_i^p$ and $\mathbf{x}_j^q$, the residual vector is computed as

$$e_{ij}(\mathbf{x}_i^p, \mathbf{x}_j^q) \triangleq \log\left((\tilde{z}_{ij}^{pq})^{-1} \boxplus (\mathbf{x}_i^p)^{-1} \boxplus \mathbf{x}_j^q\right), \quad (8)$$

with $\log(\cdot)$ from (4), and where the measurement $\tilde{z}_{ij}^{pq}$ is treated as a parameter of the function e_{ij} rather than an input.

For each $p \in \mathcal{R}$, robot p stores the vector of pose estimates

$$X_p \triangleq (\mathbf{x}_1^p, \mathbf{x}_2^p, \dots, \mathbf{x}_{N_p}^p) \in \mathcal{M}^{N_p}$$

onboard, and robot p seeks to compute optimal values for these estimates by minimizing residuals of the form of (8) along edges in a given pose graph. We also define $\Omega_{ij}^{pq} \triangleq (\Sigma_{ij}^{pq})^{-1}$, and then the weighted norm of the residuals is

$$\|e_{ij}(\mathbf{x}_i^p, \mathbf{x}_j^q)\|_{\Omega_{ij}^{pq}}^2 \triangleq e_{ij}(\mathbf{x}_i^p, \mathbf{x}_j^q)^\top \Omega_{ij}^{pq} e_{ij}(\mathbf{x}_i^p, \mathbf{x}_j^q). \quad (9)$$

B. Local Cost Functions

Robot p has a local cost on the error in its estimates of its trajectory. This cost is denoted f_p , and it contains an unshared cost h_p over robot p 's intra-robot edges, as well as a shared cost $s_{p \leftrightarrow q}$ for each $q \in S(p)$ over its inter-robot edges with a neighboring robot q . The cost f_p takes the form

$$f_p(X_p, X_{S(p)}) = h_p(X_p) + \frac{1}{2} \sum_{q \in S(p)} s_{p \leftrightarrow q}(X_p, X_q), \quad (10)$$

where $S(p)$ is from (5). The factor of $\frac{1}{2}$ is used to avoid double-counting edges when all costs are summed. We define $X_{S(p)} \triangleq \{X_q\}_{q \in S(p)}$ to be the set of trajectory estimates of all neighbors of robot p in the pose graph, and

$$h_p(X_p) = \frac{1}{2} \sum_{(i,j) \in E_p^{\text{un}}} \|e_{ij}(\mathbf{x}_i^p, \mathbf{x}_j^p)\|_{\Omega_{ij}^{pp}}^2 \quad (11)$$

$$s_{p \leftrightarrow q}(X_p, X_q) = \frac{1}{2} \sum_{((a,i),(b,j)) \in E_{pq}^{\text{sh}}} \|e_{ij}(\mathbf{x}_i^a, \mathbf{x}_j^b)\|_{\Omega_{ij}^{ab}}^2 \quad (12)$$

where $\|e_{ij}(\cdot, \cdot)\|_{\Omega_{ij}^{ab}}$ is from (9). Each inter-robot edge appears in the local costs of the two robots connected by it.

C. Global Optimization Problem Statement

The robots' objective is to find trajectory estimates that minimize their costs. Let $\mathcal{N} \triangleq \sum_{p=1}^R N_p$ denote the total number of vertices in the multi-robot pose graph $\mathcal{G}$. Define $\mathcal{X} \triangleq (X_1, \dots, X_R) \in \mathcal{M}^{\mathcal{N}}$ as the set of all vertex estimates in the multi-robot system. Then, the robots' global cost is

$$\mathcal{F}(\mathcal{X}) = \sum_{p=1}^R h_p(X_p) + \frac{1}{4} \sum_{p=1}^R \sum_{q \in S(p)} s_{p \leftrightarrow q}(X_p, X_q). \quad (13)$$

The values of $\mathcal{F}$ are invariant to rigid transformations of all poses, which makes the minimization of $\mathcal{F}$ ill-posed. Therefore, one must “anchor” at least one vertex by setting $\mathbf{x}_a^r \triangleq \mathbf{1}$ for some $(r, a) \in \mathcal{V}$, and we assume this has been done. The rest of the paper solves the following problem.

Problem 1. Using $R \in \mathbb{N}$ robots that asynchronously compute and communicate, find $\mathcal{X}^* \in \arg\min_{\mathcal{X} \in \mathcal{M}^{\mathcal{N}}} \mathcal{F}(\mathcal{X})$, where $\mathcal{F}$ is from (13).

Problem 1 is a non-convex least-squares problem on a product of Riemannian manifolds. We will solve it in a distributed manner, where, for all $p \in \mathcal{R}$, robot p iteratively computes new values of its trajectory estimate X_p using only communications with neighbors whose indices are in $S(p)$.

IV. PROPOSED ALGORITHM

In this section, we present the distributed asynchronous Riemannian Levenberg-Marquardt (RLM) algorithm that we use to solve Problem 1. These developments draw in part from [28], which developed a centralized RLM algorithm. To the best of our knowledge, the algorithm we present is the first distributed asynchronous RLM algorithm in the literature, and it may be of independent interest.

The proposed algorithm can be viewed as an asynchronous randomized block-coordinate method on the PUDQ product manifold $\mathcal{M}^N$, though each robot solves a regularized Riemannian Levenberg-Marquardt system to find a search direction, rather than using a gradient. For each $p \in \mathcal{R}$, the trajectory X_p forms one block of the global optimization variable. We use a gossip approach [29] to randomly select an “active” robot to perform a computation. At iteration k , only the block associated with the active robot is updated, while the other robots’ blocks remain unchanged. Each block update is computed using a damped Riemannian Levenberg-Marquardt step and may use outdated estimates of neighboring robots’ poses because of asynchronous communications.

Assumption 1 (Weak connectivity). *The multi-robot pose graph $\mathcal{G}$ is weakly connected.*

This assumption is mild because a non-weakly connected pose graph is comprised by some number of disjoint weakly connected components, and each one can be separately optimized as a weakly connected pose graph.

A. Riemannian Exponential and Logarithm

The algorithm we develop requires the Riemannian exponential for $\mathcal{M}$, which maps tangent vectors along geodesics to points on $\mathcal{M}$. Let $\psi(\mathbf{x}_r) \triangleq \arctan(x_1, x_0)$ be the map that computes the angle associated with a point $\mathbf{x}_r$ on the unit circle, and let $\psi^{-1}(\alpha) \triangleq [\cos(\alpha), \sin(\alpha)]^\top$ be its inverse. Then, given a pose $\mathbf{x} = [\mathbf{x}_r^\top, \mathbf{x}_d^\top]^\top \in \mathcal{M}$ and a tangent vector $\mathbf{s} = [\mathbf{s}_r^\top, \mathbf{s}_d^\top]^\top \in \mathcal{T}_{\mathbf{x}}\mathcal{M}$, the Riemannian exponential map at $\mathbf{x}$ is $\text{Exp}_{\mathbf{x}} : \mathcal{T}_{\mathbf{x}}\mathcal{M} \rightarrow \mathcal{M}$, where

$$\text{Exp}_{\mathbf{x}}(\mathbf{s}) \triangleq \begin{bmatrix} \mathbf{x}_r \otimes \psi^{-1}((\mathbf{x}_r^{-1} \otimes \mathbf{s}_r)_2) \\ \mathbf{x}_d + \mathbf{s}_d \end{bmatrix}, \quad (14)$$

where $\otimes$ is the Hamilton product and $(\cdot)_2$ outputs the second element of a vector. This map will be used in Section IV to generate iterates of an optimization algorithm that always stay on $\mathcal{M}$. This map extends to the product manifold $\mathcal{M}^N$. For any $\mathbf{s} = \text{vec}((\mathbf{s}_i)_{i=1}^N) \in \mathcal{T}_{\mathbf{x}}\mathcal{M}^N$, we define

$$\text{Exp}_X(\mathbf{s}) = \text{vec}\left((\text{Exp}_{\mathbf{x}_i}(\mathbf{s}_i))_{i=1}^N\right), \quad (15)$$

where $X = \text{vec}((\mathbf{x}_i)_{i=1}^N) \in \mathcal{M}^N$ and $\text{Exp}_{\mathbf{x}_i}(\cdot)$ from (14).

B. Timing of Computations and Communications

We model robots’ computations using a randomized gossip algorithm [14], [29], where each robot has an onboard Poisson clock that ticks at rate $\omega > 0$. If a robot’s clock ticks at times $\{\nu_1, \nu_2, \dots\}$, then the inter-arrival times are i.i.d. exponential: $\mathbb{P}[\nu_{i+1} - \nu_i \leq a] = 1 - e^{-\omega a}$ for $i \in \mathbb{N}$ and $a \geq 0$. We define a global iteration counter $k \in \mathbb{N} \cup \{0\}$ that increments at every tick of any robot’s clock. We use k only for analysis, and it is not known by the robots. This use of Poisson clocks ensures that the clock of a uniformly randomly selected robot ticks at iteration k .

Each robot $p \in \mathcal{R}$ stores an estimate of its own trajectory, which is denoted $X_p(k)$ at iteration k , and it also stores a set of estimates of its neighbors’ trajectories, denoted

$\hat{X}_{S(p)}(k) \triangleq \{\hat{X}_q^p(k)\}_{q \in S(p)}$ at iteration k . Here, $\hat{X}_q^p(k)$ contains robot p ’s onboard estimates of robot q ’s poses. These estimates are sent by robot q to robot p , and, due to asynchronous communications, they may contain old values of robot q ’s shared poses. For brevity, we denote all of robot p ’s onboard trajectory estimates at iteration k by

$$D_{p_k} \triangleq (X_p(k), \hat{X}_{S(p)}(k)). \quad (16)$$

Remark 1. *For ease of notation, we will write that robot q sends all elements of X_q to robot p when communicating, though in practice robot q only needs to send robot p estimates of any shared poses that it has with robot p .*

Each robot $p \in \mathcal{R}$ computes new estimates of its own trajectory X_p over time, and each new estimate overwrites the old estimate it had onboard. After that, robot p broadcasts its shared poses to all robots q with $q \in S(p)$, though these communications may take some time to arrive. Reciprocally, when robot p receives a communication from a neighbor $q \in S(p)$, robot p overwrites its local estimate $\hat{X}_q^p$ with the new value it receives. If robot p ’s Poisson clock ticks at iteration k , then it computes $X_p(k+1)$ using its current trajectory $X_p(k)$ and its local (possibly outdated) copies of estimates of shared poses of its neighbors in $\hat{X}_{S(p)}(k)$. This local computation does not change $\hat{X}_{S(p)}(k)$, which remains constant until new shared poses are received from neighbors.

We define $\ell_q^p(k)$ to be the iteration at which robot q first computed the estimates of its shared poses that robot p has onboard at time k in $\hat{X}_q^p(k)$. Then, $\hat{X}_q^p(k) = X_q(\ell_q^p(k))$. We have $\ell_q^p(k) \leq k$ by definition, and we assume the following.

Assumption 2 (Bounded Communication Delays). *For all robots $p, q \in \mathcal{R}$ and all iterations k , there exists $B \in \mathbb{N}$ such that $k - \ell_q^p(k) \leq B$.*

Assumption 2 ensures that the lengths of all communication delays are bounded by B , though B may be large.

C. The RLM Subproblem for Computations

We next describe each robot’s computations, and Algorithm 1 below gives an overview of all robots’ behaviors.

By the Poisson clock definition in Section IV-B, let p_k denote the robot whose clock ticks at iteration k . This robot updates its local trajectory estimate $X_{p_k}(k)$. To compute $X_{p_k}(k+1)$, it seeks a direction $s_{p_k}(k) \in \mathcal{T}_{X_{p_k}(k)}\mathcal{M}^{N_{p_k}}$ to move its estimate in. This direction is found by minimizing

$$\theta_{p_k}(s) \triangleq 2f_{p_k} + 2s^\top g_{p_k} + s^\top H_{p_k}s + \lambda_k \|s\|^2, \quad (17)$$

where $f_{p_k} \triangleq f_{p_k}(D_{p_k})$ is the local cost from (10), D_{p_k} is from (16), $g_{p_k} \triangleq \text{grad} f_{p_k}(D_{p_k})$ is the Riemannian gradient of f_{p_k} at X_{p_k} with respect to its first argument, and $H_{p_k} \triangleq H_{p_k}(D_{p_k})$ is the standard Riemannian Gauss-Newton (RGN) Hessian approximation. Definitions of g_{p_k} and H_{p_k} are included in Appendix B. Furthermore, λ_k in (17) is the damping parameter, which we set as

$$\lambda_k = 2\mu_{p_k} f_{p_k}, \quad (18)$$

for some robot-specified constant $\mu_{p_k} > 0$. Using (17), robot p_k computes the search direction s_{p_k} , defined as

$$s_{p_k} \triangleq \operatorname{argmin}_{s \in T_{X_{p_k}(k)} \mathcal{M}^{N_{p_k}}} \theta_{p_k}(s). \quad (19)$$

Using [28, Prop. 1], the optimal s_{p_k} is found by solving

$$(H_{p_k} + \lambda_k I) s_{p_k} = g_{p_k}. \quad (20)$$

Because H_{p_k} is positive semi-definite by construction, the operator $(H_{p_k} + \lambda_k I)$ is positive definite for any $\lambda_k > 0$, which implies the existence of a unique solution s_{p_k} to (20).

To help ensure convergence, we use a stepsize to scale each step taken. We use $\gamma_k \in (0, 1]$ to denote the stepsize used by robot p_k at iteration k , and then robot p_k 's tentative update, denoted $X_{p_k}^{\text{ten}}(s_{p_k})$, is computed as

$$X_{p_k}^{\text{ten}}(s_{p_k}) = \operatorname{Exp}_{X_{p_k}(k)}(-\gamma_k s_{p_k}), \quad (21)$$

where s_{p_k} is from (20) and $\operatorname{Exp}_{X_{p_k}(k)}$ is from (15). This stepsize is used to enforce certain sufficient conditions in the forthcoming convergence results in Section V, and the conditions we impose on γ_k will be stated in Section V.

The goal of the step in (21) is for robot p_k 's updated trajectory estimate $X_{p_k}(k+1)$ to produce a lower value of the cost f_{p_k} than the prior trajectory estimate $X_{p_k}(k)$. Since s_{p_k} is computed in (19) by minimizing θ_{p_k} , robot p_k evaluates the extent to which the minimization of θ_{p_k} also provides a reduction in f_{p_k} . This evaluation is done by computing

$$\rho_k \triangleq \frac{f_{p_k}(D_{p_k}) - f_{p_k}(X_{p_k}^{\text{ten}}(s_{p_k}), \hat{X}_{S(p_k)}(k))}{\frac{1}{2}(\theta_{p_k}(0) - \theta_{p_k}(-\gamma_k s_{p_k}))}, \quad (22)$$

where $X_{p_k}^{\text{ten}}(s_{p_k})$ is from (21) and D_{p_k} is from (16). The ‘‘gain ratio’’ ρ_k is equal to the reduction in f_{p_k} provided by executing (21) divided by the reduction in θ_{p_k} provided by executing (21). For a choice of threshold $\eta \in (0, 1)$, if $\rho_k \geq \eta$, then the step taken in (21) is accepted, and robot p_k sets $X_{p_k}(k+1) = X_{p_k}^{\text{ten}}(s_{p_k})$. If $\rho_k < \eta$, then the step is rejected, and the value of $X_{p_k}^{\text{ten}}(s_{p_k})$ from (21) is discarded. Next, λ_k is increased, which changes θ_{p_k} , and a new direction s_{p_k} is found by solving (19) with the new θ_{p_k} . Then, $X_{p_k}^{\text{ten}}(s_{p_k})$ is computed again via (21) and used to compute a new value of ρ_k . This process is repeated for iteration k until a step is accepted. Theorem 1 below bounds the maximum number of increases in λ_k that can occur.

D. Global View of Algorithm 1

Algorithm 1 gives a global view of the distributed asynchronous RLM algorithm, which is used only for analysis; the robots executing it do not need to know any information besides what they store onboard. As noted in Section IV-B, with identical onboard Poisson clock rates, each robot $p_k \in \mathcal{R}$ is selected uniformly randomly at iteration k . Communications are asynchronous, subject only to Assumption 2.

Algorithm 1 Distributed RLM (Global View for Analysis)

Require:

Initial iterate $\mathcal{X}(0) = (X_p(0))_{p=1}^R$ given to all robots,
 RLM acceptance threshold $\eta \in (0, 1)$,
 Damping parameters $\mu_{\min} > 0, \beta > 1$, set $\mu_0 = \mu_{\min}$
 Stepsizes $\{\gamma_k\}_{k \geq 0}$ with $\gamma_k \in (0, 1]$ for all $k \geq 0$
 Number of iterations $K \in \mathbb{N}$

- 1: **for** $k = 0, 1, 2, \dots, K$ **do**
- 2: Select robot $p_k \in \mathcal{R}$ uniformly at random
- 3: Compute $f_{p_k}, g_{p_k}, H_{p_k}$ using (10), (25), and (28).
- 4: $\mu_k \leftarrow \mu_0$
- 5: **while** true **do** ▷ Inner damping loop
- 6: Compute λ_k via (18)
- 7: Compute the search direction s_{p_k} using (20)
- 8: Compute gain ratio ρ_k using (22)
- 9: **if** $\rho_k \geq \eta$ **then** ▷ Accept
- 10: $X_{p_k}(k+1) \leftarrow \operatorname{Exp}_{X_{p_k}(k)}(-\gamma_k s_{p_k})$
- 11: **break**
- 12: **else** ▷ Reject: increase damping
- 13: $\mu_k \leftarrow \beta \mu_k$
- 14: **for** $q \in \mathcal{R}$ **do**
- 15: **for** $r \in \mathcal{R} \setminus \{q\}$ **do**
- 16: **if** Robot q receives X_r from robot r **then**
- 17: $\hat{X}_r^q(k) \leftarrow X_r(\ell_r^q(k))$
- 18: $\hat{X}_{S(q)}(k+1) \leftarrow \hat{X}_{S(q)}(k)$

V. CONVERGENCE ANALYSIS

This section analyzes the convergence of Algorithm 1 by proving finite termination of the inner damping loop and deriving a rate of convergence toward a stationary point of $\mathcal{F}$.

Theorem 1 (Well-definedness of the damping loop). *At iteration k of the outer loop of Algorithm 1, the inner damping loop in lines 5-13 terminates in at most $J_{\max} = \left\lceil \frac{1}{\log(\beta)} \log \left(\frac{\kappa}{2\mu_{\min} f_{p_k}(D_{p_k})} \right) \right\rceil$ steps, where $\lceil \cdot \rceil$ denotes the ceiling function, $\mu_{\min}$ is from Algorithm 1, and κ is from Lemma 14.*

Proof. See Appendix E. ■

Now we derive a global rate of convergence, which is our main result. In it, we use $\mathcal{X}(k) = (X_1^1(k), X_2^2(k), \dots, X_R^R(k))$, which contains all robots' estimates of their own poses at iteration $k \in \mathbb{N} \cup \{0\}$.

Theorem 2 (Finite-Time Global Convergence Rate). *Suppose Assumptions 1 and 2 hold. If the step size $\gamma_k \in (0, 1]$ is chosen so that $0 < \gamma_k < \frac{2\lambda_k}{(L_g + \Delta L_E \delta B)}$, then Algorithm 1 produces iterates that satisfy*

$$\min_{k \in \{0, \dots, K-1\}} \mathbb{E} [\|\operatorname{grad} \mathcal{F}(\mathcal{X}(k))\|^2] \leq \frac{\Upsilon \operatorname{RF}(\mathcal{X}(0))}{K \sigma_{\min}},$$

where $\operatorname{grad} \mathcal{F}(\mathcal{X}(k))$ is the Riemannian gradient of the objective function $\mathcal{F}$ from (13), $\Upsilon \triangleq 2\zeta^2 + \frac{1}{2}(\Delta L_E \delta B)^2$, $\sigma_{\min} \triangleq \min_{k \in \{0, 1, \dots, K-1\}} \sigma_k > 0$, $\sigma_k \triangleq \gamma_k \lambda_k - \gamma_k^2 \left(\frac{L_g}{2} + \right.$

$\frac{\Delta L_E \delta B}{2}$), R is the number of robots, and the expectation is taken over the robots' Poisson clocks.

Proof. See Appendix G. ■

Given any tolerance $\epsilon_g > 0$, one can use Theorem 2 to compute a maximum number of iterations K needed to find an iterate that satisfies $\mathbb{E} [\|\text{grad } \mathcal{F}(\mathcal{X}(k))\|^2] \leq \epsilon_g$.

VI. SIMULATION RESULTS

This section presents simulation results for Algorithm 1.

A. Simulation Setup and Noise Modeling

We generated 15 pose graph datasets with three ground-truth models by corrupting the edge measurements of each with five different levels of noise drawn from the Lie-theoretic noise model in (7), thereby introducing anisotropic uncertainty. For each edge $((p, i), (q, j))$, the measurement noise covariance Σ_{ij}^{pq} was sampled from a Wishart [30] distribution $W_d(V, n)$, with $d = 3$ and $n = 10$. The scale matrix was set to $V = \sigma_w \Sigma_w$, where $\Sigma_w \triangleq \mathbb{1}_{3 \times 3} + \text{diag}([u_1, u_2, u_3])$, $\mathbb{1}_{3 \times 3}$ is the 3×3 matrix of ones, each u_i is uniformly sampled from $(0, 1]$, and σ_w is a scalar.

By varying σ_w between 10^{-5} and 10^{-2} , we produced random positive-definite covariance matrices with expected value $\mathbb{E}[\Sigma_{ij}^{pq}] = 10\sigma_w \Sigma_w$. Across the 15 datasets that were generated, the expected translation uncertainty has a standard deviation of 7.26 mm to 22.9 cm. It also gives the expected rotation uncertainty a standard deviation of 0.41° to 12.81° . The upper bounds of these noise levels are larger than the noise levels used with the ASAPP framework in [14] for distributed asynchronous PGO, which are 5 cm and 2° .

For all simulations, the first pose was anchored at the identity ($\mathbf{x}_1 \triangleq \mathbf{1}$), and although Algorithm 1 does not require a specific initialization, all methods were run with chordal initialization, which is typically used with SE-Sync and Cartan-Sync. All simulations were run in C++/ROS for three datasets: a synthesized Grid1000 dataset (1000 vertices, 1250 edges), the Intel dataset [31] (1729 vertices, 2512 edges), and the M3500 dataset [31] (3500 vertices, 5598 edges).

B. Distributed and Asynchronous Implementation

We partition each graph across five simulated robots. In implementation, every robot runs as an independent ROS 2 node. This distributed implementation separates local optimization and communication into two concurrent processes.

As noted in Section IV-B, each robot triggers its local optimization via an independent Poisson clock, and we use the trigger rate $\omega = 1000 \text{ s}^{-1}$. In all experiments, we set the initial RLM damping factor to $\mu_{\min} = 0.1$, the step acceptance threshold to $\eta = 0.15$, and the damping increase factor to $\beta = 5$. As in existing work, e.g., [14] and references therein, we set stepsizes empirically, and we use $\gamma_k = 1$ for all iterations k . Communication runs concurrently with local optimization, and each robot periodically broadcasts its current pose estimates using ROS 2. We denote the communication period between consecutive broadcasts by T_c .

Each robot can execute multiple optimization steps between two communications, which causes its computations to rely on potentially outdated estimates of its neighbors' poses. We emphasize that the communication period T_c is different from the delay bound B in Assumption 2. The value of B is in terms of ticks of k in Algorithm 1, whereas T_c is measured in seconds. In practice, the robots' computation rate, communication period, computation time, and network latency jointly determine the delay bound B .

C. Baseline Methods

We compare Algorithm 1 to two centralized solvers, SE-Sync [10] and Cartan-Sync [11], and two distributed asynchronous solvers, DC2-PGO [32] and ASAPP [14]. All algorithms are evaluated on identical pose graphs, ground-truth trajectories, and initial estimates, with distributed methods deployed across identical five-robot pose graphs.

D. Evaluation Metrics

Solutions computed by each algorithm were evaluated using the root-mean-square relative pose error (RPE) metric, which quantifies the total geometric deformation of the final estimated poses $(\hat{\mathbf{x}}_i)_{i=1}^N$ relative to the ground truth poses $(\mathbf{x}_i^\circ)_{i=1}^N$. RPE provides a natural evaluation metric for PGO because the measurements in a pose graph describe relative transformations between connected poses. Therefore, RPE directly evaluates how accurately an estimated trajectory recovers these relative relationships, rather than depending on the absolute position or orientation of the entire trajectory in a global reference frame.

The Lie-theoretic RPE (RPE-L) is defined using $\log$ from (4) as $\text{RPE-L} \triangleq \left(\frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \left\| \log(\hat{\mathbf{z}}_{ij}^{-1} \boxplus \mathbf{z}_{ij}^\circ) \right\|^2 \right)^{\frac{1}{2}}$, where $\hat{\mathbf{z}}_{ij} \triangleq \hat{\mathbf{x}}_i^{-1} \boxplus \hat{\mathbf{x}}_j$ and $\mathbf{z}_{ij}^\circ \triangleq (\mathbf{x}_i^\circ)^{-1} \boxplus \mathbf{x}_j^\circ$. We define the Euclidean RPE (RPE-E) using the translational vectors $\hat{\mathbf{t}}_{ij}$ and rotational components $\hat{\theta}_{ij}$ of the robots' pose estimates as $\text{RPE-E} \triangleq \left(\frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \left(\left\| \hat{\mathbf{t}}_{ij} - \mathbf{t}_{ij}^\circ \right\|^2 + d(\hat{\theta}_{ij}, \theta_{ij}^\circ)^2 \right) \right)^{\frac{1}{2}}$, where $d(\theta_1, \theta_2)$ is the minimal angle between θ_1 and θ_2 .

To ensure a fair and comprehensive benchmark, we evaluate every algorithm using both RPE-L and RPE-E. While RPE-L captures the intrinsic geometric error on the PUDQ manifold, RPE-E breaks the error down into familiar Cartesian translation and heading angles. Reporting both metrics ensures a fair comparison by considering both trajectory error measured on the manifold $\text{SE}(2)$ and in standard Euclidean space, which are both of interest.

E. Estimation Accuracy under Anisotropic Noise

Table I compares the asymptotic accuracy of all algorithms across all 15 pose graphs. Cartan-Sync and SE-Sync were run synchronously, and DC2-PGO, ASAPP, and Algorithm 1 were run asynchronously with a communication period of $T_c = 0.2\text{s}$. We see that Algorithm 1 produces lower values of RPE-L and RPE-E than all other algorithms. For the other distributed asynchronous algorithms, namely ASAPP and DC2-PGO, Figure 2 shows that, across all datasets, Algorithm 1 produces RPE-L values that are 3.6% to 21%

TABLE I

VALUES OF RPE ACROSS STANDARD BENCHMARK PGO DATASETS UNDER ANISOTROPIC NOISE. BOLD INDICATES THE LOWEST ERROR, WITH PERCENT REDUCTION RELATIVE TO THE MOST ACCURATE BASELINE IN EACH CASE SHOWN IN PARENTHESES.

Dataset	Noise (σ_w)	Cartan-Sync		SE-Sync		ASAPP		DC2-PGO		Alg. 1 (Ours) (% Reduction)	
		RPE-L	RPE-E	RPE-L	RPE-E	RPE-L	RPE-E	RPE-L	RPE-E	RPE-L	RPE-E
Grid1000	$1.0 \cdot 10^{-5}$	0.0061	0.0123	0.0061	0.0123	0.0061	0.0123	0.0061	0.0123	0.0054 (-11.5%)	0.0108 (-12.2%)
	$5.6 \cdot 10^{-5}$	0.0147	0.0294	0.0147	0.0294	0.0147	0.0294	0.0147	0.0295	0.0129 (-12.2%)	0.0259 (-11.9%)
	$3.2 \cdot 10^{-4}$	0.0353	0.0707	0.0353	0.0707	0.0353	0.0707	0.0354	0.0708	0.0311 (-11.9%)	0.0622 (-12.0%)
	$1.8 \cdot 10^{-3}$	0.0792	0.1584	0.0792	0.1584	0.0792	0.1584	0.0793	0.1587	0.0709 (-10.5%)	0.1419 (-10.4%)
	$1.0 \cdot 10^{-2}$	0.1947	0.3885	0.1947	0.3885	0.2051	0.4092	0.1958	0.3906	0.1800 (-7.6%)	0.3592 (-7.5%)
Intel	$1.0 \cdot 10^{-5}$	0.0056	0.0112	0.0056	0.0112	0.0056	0.0112	0.0056	0.0113	0.0049 (-12.5%)	0.0099 (-11.6%)
	$5.6 \cdot 10^{-5}$	0.0131	0.0263	0.0131	0.0263	0.0131	0.0263	0.0131	0.0263	0.0119 (-9.2%)	0.0238 (-9.5%)
	$3.2 \cdot 10^{-4}$	0.0316	0.0632	0.0316	0.0632	0.0316	0.0632	0.0328	0.0657	0.0290 (-8.2%)	0.0580 (-8.2%)
	$1.8 \cdot 10^{-3}$	0.0740	0.1479	0.0740	0.1479	0.0740	0.1479	0.0758	0.1516	0.0676 (-8.6%)	0.1351 (-8.7%)
	$1.0 \cdot 10^{-2}$	0.1776	0.3543	0.1776	0.3543	0.1760	0.3511	0.1842	0.3675	0.1697 (-3.6%)	0.3386 (-3.6%)
M3500	$1.0 \cdot 10^{-5}$	0.0054	0.0108	0.0054	0.0108	0.0054	0.0108	0.0054	0.0109	0.0043 (-20.4%)	0.0087 (-19.4%)
	$5.6 \cdot 10^{-5}$	0.0131	0.0263	0.0131	0.0263	0.0132	0.0264	0.0133	0.0267	0.0105 (-19.8%)	0.0211 (-19.8%)
	$3.2 \cdot 10^{-4}$	0.0310	0.0621	0.0310	0.0621	0.0314	0.0628	0.0315	0.0630	0.0248 (-20.0%)	0.0496 (-20.1%)
	$1.8 \cdot 10^{-3}$	0.0729	0.1458	0.0729	0.1458	0.0734	0.1469	0.0738	0.1476	0.0592 (-18.8%)	0.1185 (-18.7%)
	$1.0 \cdot 10^{-2}$	0.1745	0.3484	0.1745	0.3484	0.1779	0.3558	0.1785	0.3563	0.1428 (-18.2%)	0.2857 (-18.0%)

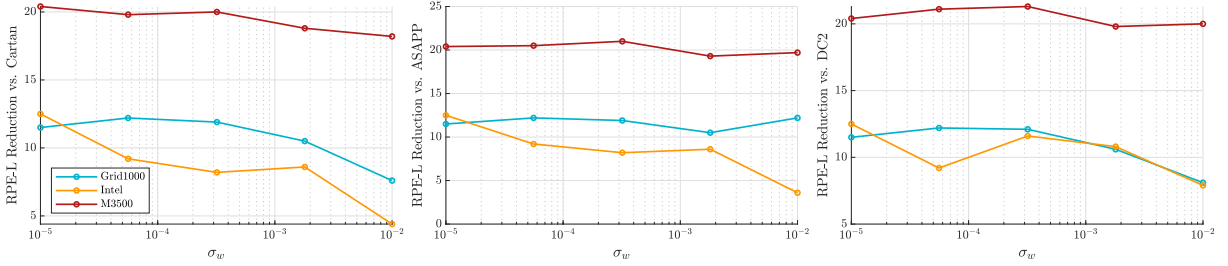


Fig. 2. Percentage reduction in RPE-L achieved by Algorithm 1 (ours) compared across the Grid1000, Intel, and M3500 datasets against centralized and distributed baselines: Cartan-Sync/SE-Sync (left), ASAPP (middle), and DC2-PGO (right) across varying anisotropic noise levels σ_w .

TABLE II

EFFECT OF COMMUNICATION PERIOD T_c ON ASYMPTOTIC RPE
VALUES PRODUCED BY ALGORITHM 1.

Dataset	T_c [s]	Low Noise $\sigma_w = 10^{-5}$		High Noise $\sigma_w = 10^{-2}$	
		RPE-L	RPE-E	RPE-L	RPE-E
Intel	0.05	0.0050	0.0101	0.1697	0.3385
	0.10	0.0050	0.0101	0.1693	0.3378
	0.20	0.0050	0.0101	0.1695	0.3381
	0.50	0.0050	0.0101	0.1698	0.3388
	1.00	0.0051	0.0102	0.1701	0.3392
M3500	0.05	0.0044	0.0088	0.1433	0.2862
	0.10	0.0044	0.0088	0.1432	0.2860
	0.20	0.0044	0.0088	0.1433	0.2862
	0.50	0.0044	0.0088	0.1433	0.2862
	1.00	0.0044	0.0088	0.1438	0.2871

lower than ASAPP and 7.9% to 21.3% lower than DC2-PGO. We omit RPE-E values from Figure 2 for clarity, though they behave similarly. These improvements in estimation error are largest for the M3500 dataset, which is the largest graph we evaluated. Although Algorithm 1 is both decentralized and asynchronous, it outperforms centralized solvers and other distributed solvers on PGO problems under anisotropic noise.

F. Effect of Communication Period

We next evaluate the sensitivity of Algorithm 1 to changes in the communication period T_c while keeping other parameters fixed. We use the values $T_c \in \{0.05s, 0.10s, 0.20s, 0.50s, 1.00s\}$, which correspond to

communication rates of 20 Hz, 10 Hz, 5 Hz, 2 Hz, and 1 Hz, respectively. Larger values of T_c reduce the inter-robot communication frequency, which increases the potential outdatedness of the robots' onboard information.

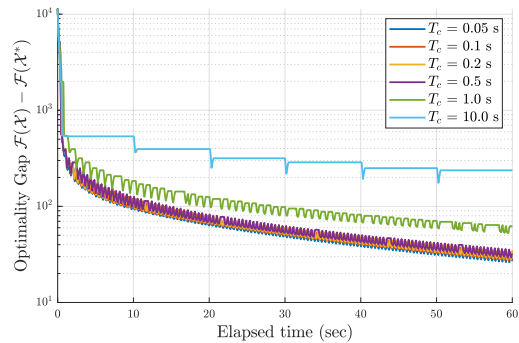


Fig. 3. The impact of communication delay on the optimality gap seen by Algorithm 1 on the M3500 dataset with $\sigma_w = 10^{-2}$ over 60 seconds. Even under a $20\times$ change in communication delay, the optimality gap rapidly decreases initially, followed by a decrease at a slower rate thereafter.

Table II shows that for the Intel data set, the final estimation accuracy of Algorithm 1 remains nearly unchanged across all tested communication periods. Under low noise, RPE-L varies only from 0.0050 to 0.0051 and RPE-E varies from 0.0101 to 0.0102. Under high noise, RPE-L ranges

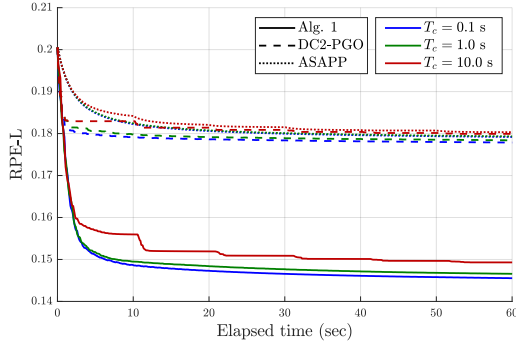


Fig. 4. The impact of varying the communication delay on RPE-L convergence over time for Algorithm 1, DC2-PGO, and ASAPP. Each algorithm processed the M3500 dataset with $\sigma_w = 10^{-2}$ for 60 seconds. We see that increased communication delays lead to slower convergence in general, but Algorithm 1 quickly achieves superior RPE-L in all cases.

from 0.1693 to 0.1701 and RPE-E ranges from 0.3378 to 0.3392. These values are quite close to those in Table I, which indicates that different communication periods have minimal impact on the accuracy of the robots' final estimates. Even under longer delays, most of the decrease in the cost $\mathcal{F}$ is attained in the first few seconds of running Algorithm 1, which is shown in Figure 3. In it, we see that all communication periods lead to an initial rapid decrease in $\mathcal{F}$, followed by a slower decrease after that.

For M3500, Table II shows similarly stable low-noise results (RPE-L $\approx$ 0.0044, RPE-E $\approx$ 0.0088) and minimal high-noise variation (RPE-L from 0.1432 to 0.1438; RPE-E from 0.2860 to 0.2871) across all T_c values.

Figure 4 compares RPE-L against ASAPP [14]. For visual clarity, we present only a plot of RPE-L, though RPE-E results are similar. Figure 4 shows that Algorithm 1 produces both a faster initial decrease in RPE-L and a lower value of it over the runtimes of these algorithms, and these improvements hold for communication delays spanning three orders of magnitude. Within 60 seconds, all curves drop below 0.15, and they asymptotically converge to the high-noise value of 0.1438 from Table II.

VII. CONCLUSION

We presented a distributed asynchronous Riemannian Levenberg-Marquardt algorithm for pose graph optimization over planar unit dual quaternions. We then proved that it converges to first-order critical points under bounded communication delays. Simulations showed that, on problems with anisotropic noise, our method achieves lower estimation error than state-of-the-art synchronous solvers and state-of-the-art distributed asynchronous solvers. Future work will extend this framework to 3D pose graphs.

REFERENCES

- [1] D. S. Drew, "Multi-agent systems for search and rescue applications," *Current Robotics Reports*, vol. 2, no. 2, pp. 189–200, 2021.
- [2] F. Maresca, F. Devoti, G. Encinas-Lago, V. Sciancalepore, J. Paradells, and X. Costa-Perez, "Localization-driven multi-robot exploration for indoor search and rescue," *IEEE Open J. Commun. Soc.*, vol. 7, pp. 421–441, 2026.

- [3] J.-P. de la Croix, F. Rossi, R. Brockers, D. Aguilar, K. Albee, E. Boroson, A. Cauligi, J. Delaune, R. Hewitt, D. Kogan *et al.*, "Multi-agent autonomy for space exploration on the cadre lunar technology demonstration," in *2024 IEEE Aerosp. Conf.* IEEE, 2024, pp. 1–14.
- [4] C. Zheng, Y. Bao, and C. Wei, "Multi-robot autonomous cooperative exploration in unknown environment based on region assignment," in *Int. Symp. Rob. Intell. Manuf. Technol. (ISRIMT)*, 2024, pp. 119–123.
- [5] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard, "Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age," *IEEE Trans. Rob.*, vol. 32, no. 6, pp. 1309–1332, 2017.
- [6] D. Zou, P. Tan, and W. Yu, "Collaborative visual slam for multiple agents: A brief survey," *Virtual Reality & Intelligent Hardware*, vol. 1, no. 5, pp. 461–482, 2019.
- [7] K. He, R. Jia, H. Hong, N. Wang, and Y. Hu, "Ldg-cslam: Multi-robot collaborative slam based on curve analysis, normal distribution, and factor graph optimization," *J. Field Rob.*, vol. 42, no. 5, pp. 2173–2191, 2025.
- [8] P. Schmuck and M. Chli, "Ccm-slam: Robust and efficient centralized collaborative monocular simultaneous localization and mapping for robotic teams," *J. Field Rob.*, vol. 36, no. 4, pp. 763–781, 2019.
- [9] X. Zhao, S. Shao, T. Wang, C. Fang, J. Zhang, and H. Zhao, "A review of multi-robot collaborative simultaneous localization and mapping," in *Int. Conf. Unmann. Syst. (ICUS)*. IEEE, 2023, pp. 900–905.
- [10] D. M. Rosen, L. Carlone, A. S. Bandeira, and J. J. Leonard, "Se-sync: A certifiably correct algorithm for synchronization over the special euclidean group," *Int. J. Rob. Res.*, vol. 38, no. 2-3, pp. 95–125, 2019.
- [11] J. Briaies and J. Gonzalez-Jimenez, "Cartan-sync: Fast and global se (d)-synchronization," *IEEE Rob. Autom. Lett.*, vol. 2, no. 4, pp. 2127–2134, 2017.
- [12] T. Fan, H. Wang, M. Rubenstein, and T. Murphey, "Efficient and guaranteed planar pose graph optimization using the complex number representation," in *Int. Conf. Intell. Rob. Syst. (IROS)*. IEEE, 2019, pp. 1904–1911.
- [13] S. Choudhary, L. Carlone, C. Nieto, J. Rogers, H. I. Christensen, and F. Dellaert, "Distributed trajectory estimation with privacy and communication constraints: a two-stage distributed gauss-seidel approach," in *Int. Conf. Rob. Autom. (ICRA)*. IEEE, May 2016, pp. 5261–5268.
- [14] Y. Tian, A. Koppel, A. S. Bedi, and J. P. How, "Asynchronous and parallel distributed pose graph optimization," *IEEE Rob. Autom. Lett.*, vol. 5, no. 4, pp. 5819–5826, 2020.
- [15] G. Guo and P. Yi, "An asynchronous parallel algorithm framework for decentralized pose graph optimization," in *2023 3rd Int. Conf. on Comput., Control and Rob. (ICCCR)*. IEEE, 2023, pp. 158–163.
- [16] D. O. Wheeler, D. P. Koch, J. S. Jackson, T. W. McLain, and R. W. Beard, "Relative navigation: A keyframe-based approach for observable GPS-degraded navigation," *IEEE Control Syst. Mag.*, vol. 38, no. 4, pp. 30–48, 2018.
- [17] T. D. Barfoot, C. Holmes, and F. Dömbgen, "Certifiably optimal rotation and pose estimation based on the cayley map," 2024. [Online]. Available: <https://arxiv.org/abs/2308.12418>
- [18] C. Olsson, Y. Lochman, J. Malmport, and C. Zach, "Certifiably optimal anisotropic rotation averaging," 2025. [Online]. Available: <https://arxiv.org/abs/2503.07353>
- [19] M. Brossard, S. Bonnabel, and A. Barrau, "A new approach to 3d icp covariance estimation," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, p. 744–751, Apr. 2020.
- [20] D. McGann, K. Lassak, and M. Kaess, "Asynchronous distributed smoothing and mapping via on-manifold consensus admm," in *Int. Conf. Rob. Autom. (ICRA)*. IEEE, 2024, pp. 4577–4583.
- [21] J. Bird, J. Blumenkamp, and A. Prorok, "Dvm-slam: Decentralized visual monocular simultaneous localization and mapping for multi-agent systems," in *Int. Conf. Rob. Autom. (ICRA)*, 2025, pp. 1–7.
- [22] T. Fan and T. Murphey, "Majorization minimization methods for distributed pose graph optimization with convergence guarantees," in *Int. Conf. Intell. Robots Syst. (IROS)*. IEEE, 2020, pp. 5058–5065.
- [23] N. T. Dantam, "Robust and efficient forward, differential, and inverse kinematics using dual quaternions," *The International Journal of Robotics Research*, vol. 40, no. 10-11, pp. 1087–1105, 2021.
- [24] W. D. Warke, J. H. Ramos, P. Ganesh, K. M. Brink, and M. T. Hale, "Technical report: Pose graph optimization over planar unit dual quaternions: Improved accuracy with provably convergent riemannian optimization," *arXiv preprint arXiv:2404.00010*, 2024.
- [25] K. Li, J. Cox, B. Noack, and U. D. Hanebeck, "Improved pose graph optimization for planar motions using riemannian geometry on the

manifold of dual quaternions,” *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 9541–9547, 2020.

- [26] A. W. Long, K. C. Wolfe, M. J. Mashner, G. S. Chirikjian *et al.*, “The banana distribution is gaussian: A localization study with exponential coordinates,” *Rob.: Sci. Syst. VIII*, vol. 265, no. 1, 2013.
- [27] T. Barfoot and P. Furgale, “Associating uncertainty with three-dimensional poses for use in estimation problems,” *IEEE Trans. Rob.*, vol. 30, pp. 679–693, 06 2014.
- [28] S. Adachi, T. Okuno, and A. Takeda, “Riemannian levenberg-marquardt method with global and local convergence properties,” *arXiv preprint arXiv:2210.00253*, 2022.
- [29] S. Boyd, A. Ghosh, B. Prabhakar, and D. Shah, “Randomized gossip algorithms,” *IEEE Trans. Inf. Theory*, vol. 52, no. 6, pp. 2508–2530, 2006.
- [30] C. Chatfield and A. Collins, *Introduction to Multivariate Analysis*. Wiley Interscience, 2018.
- [31] L. Carlone and A. Censi, “From angular manifolds to the integer lattice: Guaranteed orientation estimation with application to pose graph optimization,” *IEEE Trans. Rob.*, vol. 30, no. 2, pp. 475–492, 2014.
- [32] Y. Tian, K. Khosoussi, D. M. Rosen, and J. P. How, “Distributed certifiably correct pose-graph optimization,” 2021. [Online]. Available: <https://arxiv.org/abs/1911.03721>
- [33] C. Udriste, *Convex functions and optimization methods on Riemannian manifolds*. Springer Science & Business Media, 2013, vol. 297.
- [34] E. Celledoni, S. Eides, B. Owren, and T. Ringholm, “Dissipative numerical schemes on riemannian manifolds with applications to gradient flows,” *SIAM J. Sci. Comput.*, vol. 40, no. 6, pp. A3789–A3806, 2018.
- [35] O. P. Ferreira, M. S. Louzeiro, and L. Prudente, “Gradient method for optimization on riemannian manifolds with lower bounded curvature,” *SIAM Journal on Optimization*, vol. 29, no. 4, pp. 2517–2541, 2019.
- [36] N. Boumal, P.-A. Absil, and C. Cartis, “Global rates of convergence for nonconvex optimization on manifolds,” *IMA J. Numer. Anal.*, vol. 39, no. 1, pp. 1–33, 2019.
- [37] R. A. Horn and C. R. Johnson, *Matrix Analysis*. Cambridge university press, 2012.
- [38] N. Boumal, *Introduction to Optimization on Smooth Manifolds*. Cambridge University Press, 2023.

APPENDIX

A. Riemannian geometry of $\mathcal{M}$ and $\mathcal{M}^N$

We briefly review the Riemannian geometry involved with our PGO problem formulation. As stated in Section II-A, we embed $\mathcal{M}$ in the ambient Euclidean space $\mathbb{R}^4$ with the Euclidean norm; thus, $\mathcal{M}$ inherits the metric $\langle u, w \rangle_{\mathbf{x}} \triangleq u^\top w$ and norm $\|u\|_{\mathbf{x}} \triangleq \|u\|_2$ for all $\mathbf{x} \in \mathcal{M}$ and $u, w \in \mathcal{T}_{\mathbf{x}}\mathcal{M}$. The manifold $\mathcal{M}^N$ admits the metric $\langle \mathcal{U}, \mathcal{W} \rangle_X = \mathcal{U}^\top \mathcal{W}$ and norm $\|\mathcal{U}\|_X \triangleq \|\mathcal{U}\|_2$ for all $X \in \mathcal{M}^N$ and $\mathcal{U}, \mathcal{W} \in \mathcal{T}_X\mathcal{M}^N$.

Given two points $\mathbf{x}, \mathbf{y} \in \mathcal{M}$, the inverse of $\text{Exp}_{\mathbf{x}}(\cdot)$ in (14) is the Riemannian logarithm map, given by

$$\text{Log}_{\mathbf{x}}(\mathbf{y}) = \begin{bmatrix} \mathbf{x}_r \otimes [0, \psi(\mathbf{x}_r^{-1} \otimes \mathbf{y}_r)]^\top \\ \mathbf{y}_d - \mathbf{x}_d \end{bmatrix}. \quad (23)$$

Similarly, for any $X, Y \in \mathcal{M}^N$, we have $\text{Log}_X(Y) = \text{vec}((\text{Log}_{\mathbf{x}_i}(\mathbf{y}_i))_{i=1}^N)$. Using (23), one can derive the associated geodesic distance, which measures the lengths of the shortest curves on $\mathcal{M}$ between points on $\mathcal{M}$. For $\mathbf{x}, \mathbf{y} \in \mathcal{M}$, it is given by $d_{\mathcal{M}}(\mathbf{x}, \mathbf{y}) = \|\text{Log}_{\mathbf{x}}(\mathbf{y})\|$, with $\text{Log}_{\mathbf{x}}(\cdot)$ from (23). The geodesic distance between $X \in \mathcal{M}^N$ and $Y \in \mathcal{M}^N$ is

$$d_{\mathcal{M}^N}(X, Y) = \sqrt{\sum_{i=1}^N d_{\mathcal{M}}(\mathbf{x}_i, \mathbf{y}_i)^2} = \sqrt{\sum_{i=1}^N \|\text{Log}_{\mathbf{x}_i}(\mathbf{y}_i)\|^2}, \quad (24)$$

with $X = \text{vec}((\mathbf{x}_i)_{i=1}^N)$, and $Y = \text{vec}((\mathbf{y}_i)_{i=1}^N)$.

B. Riemannian gradient and Hessian approximation

In (17), $g_{p_k} \triangleq \text{grad} f_{p_k}(X_{p_k}, \hat{X}_{S(p_k)})$ is the Riemannian gradient of f_{p_k} with respect to its first argument at X_{p_k} . Dropping the k subscript for convenience, it is computed as the orthogonal projection of the Euclidean gradient of f_p with respect to its first argument, denoted $\bar{\partial} f_p$, onto $T_{X_p}\mathcal{M}^{N_p}$, via

$$\text{grad} f_p(X_p, \hat{X}_{S(p)}) = \mathcal{P}_{X_p} \bar{\partial} f_p(X_p, \hat{X}_{S(p)}), \quad (25)$$

where

$$\mathcal{P}_{X_p} = \text{diag}((\mathcal{P}_{\mathbf{x}_i^p})_{i=1}^{N_p}), \quad (26)$$

along with $\mathcal{P}_{\mathbf{x}_i^p} = I_4 - \tilde{P} \mathbf{x}_i^p (\mathbf{x}_i^p)^\top \tilde{P}$, where the matrix $I_4 \in \mathbb{R}^{4 \times 4}$ is an identity matrix and $\tilde{P} \triangleq \text{diag}(1, 1, 0, 0)$. The Euclidean gradient $\bar{\partial} f_p \triangleq \bar{\partial} f_p(X_p, \hat{X}_{S(p)})$ is given by

$$\bar{\partial} f_p = \sum_{(i,j) \in E_p^{\text{un}}} \mathbf{g}_{ij}^p + \frac{1}{2} \sum_{q \in S(p)} \sum_{((a,i),(b,j)) \in E_{pq}^{\text{sh}}} \mathbf{g}_{ij}^{ab}, \quad (27)$$

where $\mathbf{g}_{ij}^p \triangleq \mathbf{g}_{ij}^p(\mathbf{x}_i^p, \mathbf{x}_j^p) = \text{vec}((g_{ij,\ell}^p)_{\ell=1}^{N_p})$, with

$$g_{ij,\ell}^p = \begin{cases} (\mathcal{A}_{ij}^p)^\top \Omega_{ij}^{pp} e_{ij}^p & i = \ell \\ (\mathcal{B}_{ij}^p)^\top \Omega_{ij}^{pp} e_{ij}^p & j = \ell \\ \mathbf{0}_{4 \times 1} & \text{otherwise} \end{cases},$$

and $e_{ij}^p \triangleq e_{ij}(\mathbf{x}_i^p, \mathbf{x}_j^p)$, $\mathcal{A}_{ij}^p \triangleq \partial e_{ij}^p / \partial \mathbf{x}_i^p$, $\mathcal{B}_{ij}^p \triangleq \partial e_{ij}^p / \partial \mathbf{x}_j^p$. In (27), $\mathbf{g}_{ij}^{ab} \triangleq \mathbf{g}_{ij}^{ab}(\mathbf{x}_i^a, \mathbf{x}_j^b) = \text{vec}((g_{ij,\ell}^{ab})_{\ell=1}^{N_p})$, where

$$g_{ij,\ell}^{ab} = \begin{cases} (\mathcal{A}_{ij}^{ab})^\top \Omega_{ij}^{ab} e_{ij}^{ab} & a = p \text{ and } i = \ell \\ (\mathcal{B}_{ij}^{ab})^\top \Omega_{ij}^{ab} e_{ij}^{ab} & b = p \text{ and } j = \ell \\ \mathbf{0}_{4 \times 1} & \text{otherwise} \end{cases},$$

where $e_{ij}^{ab} \triangleq e_{ij}(\mathbf{x}_i^a, \mathbf{x}_j^b)$, $\mathcal{A}_{ij}^{ab} \triangleq \partial e_{ij}^{ab} / \partial \mathbf{x}_i^a$, $\mathcal{B}_{ij}^{ab} \triangleq \partial e_{ij}^{ab} / \partial \mathbf{x}_j^b$. General forms for $\mathcal{A}_{ij}, \mathcal{B}_{ij}$ are derived in [24, Appendix H].

In (17), H_{p_k} approximates the Riemannian Hessian via

$$H_p(X_p, X_{S(p)}) = \mathcal{P}_{X_p} R_p(X_p, X_{S(p)}) \mathcal{P}_{X_p}, \quad (28)$$

where $\mathcal{P}_{X_p}$ is from (26), and $R_p \triangleq R_p(X_p, X_{S(p)})$ is defined as

$$R_p = \sum_{(i,j) \in E_p^{\text{un}}} R_{ij}^p + \frac{1}{2} \sum_{q \in S(p)} \sum_{((a,i),(b,j)) \in E_{pq}^{\text{sh}}} R_{ij}^{ab},$$

where $R_{ij}^p \triangleq R_{ij}^p(\mathbf{x}_i^p, \mathbf{x}_j^p)$ and $R_{ij}^{ab} \triangleq R_{ij}^{ab}(\mathbf{x}_i^a, \mathbf{x}_j^b)$, with $R_{ij}^p, R_{ij}^{ab} \in \mathbb{R}^{4N_p \times 4N_p}$. Denoting block indices $i \triangleq 4i + 1 : 4i + 4$ and $j \triangleq 4j + 1 : 4j + 4$, R_{ij}^p has four nonzero blocks, which are given by $[R_{ij}^p]_{(i,i)} = (\mathcal{A}_{ij}^p)^\top \Omega_{ij}^{pp} \mathcal{A}_{ij}^p$, $[R_{ij}^p]_{(i,j)} = [R_{ij}^p]_{(j,i)}^\top = (\mathcal{A}_{ij}^p)^\top \Omega_{ij}^{pp} \mathcal{B}_{ij}^p$, and $[R_{ij}^p]_{(j,j)} = (\mathcal{B}_{ij}^p)^\top \Omega_{ij}^{pp} \mathcal{B}_{ij}^p$, with $\mathcal{A}_{ij}^p, \mathcal{B}_{ij}^p$ defined as above. Furthermore, R_{ij}^{ab} has one nonzero block, which is either $[R_{ij}^{ab}]_{(i,i)} = (\mathcal{A}_{ij}^{ab})^\top \Omega_{ij}^{ab} \mathcal{A}_{ij}^{ab}$ if $a = p$, or $[R_{ij}^{ab}]_{(j,j)} = (\mathcal{B}_{ij}^{ab})^\top \Omega_{ij}^{ab} \mathcal{B}_{ij}^{ab}$ if $b = p$, with $\mathcal{A}_{ij}^{ab}, \mathcal{B}_{ij}^{ab}$ defined as above.

C. RLM Jacobians

For the definition of the distributed RLM algorithm we require the computation of certain Jacobians associated with Problem 1. First we define weighted residuals. Let L_{ij}^{pq} be the Cholesky factor of the information matrix Ω_{ij}^{pq} in (9) such that $\Omega_{ij}^{pq} = (L_{ij}^{pq})^\top L_{ij}^{pq}$. We define the weighted residual blocks for robot p for unshared and shared edges, respectively, as

$$\begin{aligned}\tilde{r}_{ij}^{\text{un}}(X_p) &\triangleq L_{ij}^{pq} e_{ij}(\mathbf{x}_i^p, \mathbf{x}_j^p) \text{ for all } (i, j) \in E_p^{\text{un}} \\ \tilde{r}_{ij}^{\text{sh}}(X_p, X_q) &\triangleq L_{ij}^{ab} e_{ij}(\mathbf{x}_i^a, \mathbf{x}_j^b) \text{ for all } ((a, i), (b, j)) \in E_{pq}^{\text{sh}},\end{aligned}$$

where either $a = p$ and $b = q$ or $b = p$ and $a = q$. For a given pose graph $\mathcal{G}$, let $m_p = |E_p^{\text{un}}|$ be the number of intra-robot edges belonging to robot p , and suppose they are numbered $(i_1, j_1), \dots, (i_{m_p}, j_{m_p})$. Let $n_p = \sum_{q \in S(p)} |E_{pq}^{\text{sh}}|$ be the number of inter-robot edges that connect robot p to another robot, and suppose they are numbered $(k_1, l_1), \dots, (k_{n_p}, l_{n_p})$. Then the total number of edges incident to a pose of robot p is $e_p^{\text{tot}} = m_p + n_p$. We define the map $F_p : \mathcal{M}^{N_p} \times \prod_{q \in S(p)} \mathcal{M}^{N_q} \rightarrow \mathbb{R}^{3e_p^{\text{tot}}}$ as

$$F_p(X_p, X_{S(p)}) = \begin{bmatrix} \tilde{r}_{i_1 j_1}^{\text{un}}(X_p) \\ \vdots \\ \tilde{r}_{i_{m_p} j_{m_p}}^{\text{un}}(X_p) \\ \frac{1}{\sqrt{2}} \tilde{r}_{k_1 l_1}^{\text{sh}}(X_p, X_{S(p)}) \\ \vdots \\ \frac{1}{\sqrt{2}} \tilde{r}_{k_{n_p} l_{n_p}}^{\text{sh}}(X_p, X_{S(p)}) \end{bmatrix} \in \mathbb{R}^{3e_p^{\text{tot}}}. \quad (29)$$

The local cost function for robot p may be written as

$$f_p(X_p, X_{S(p)}) = \frac{1}{2} \|F_p(X_p, X_{S(p)})\|^2,$$

where f_p is from (10). The Jacobian matrix

$$J_p = \frac{\partial F_p}{\partial X_p} \quad (30)$$

is a linear map from the tangent space $T_{X_p} \mathcal{M}^{N_p}$ to the residual space $\mathbb{R}^{3e_p^{\text{tot}}}$. Formally, it is the matrix of partial derivatives of the residual vector F_p with respect to robot p 's trajectory estimate $X_p \in \mathcal{M}^{N_p}$.

D. Technical Lemmas

This appendix presents several technical lemmas that are needed for the proofs of our main results. First, we show that although $\mathcal{M}$ is not compact, its sublevel sets are.

Lemma 1 (Compact Sublevel Sets). *Suppose Assumption 1 holds. Then the sublevel sets of $\mathcal{F}$ from Problem 1, defined as*

$$\mathcal{L}(\bar{\mathcal{X}}) \triangleq \{\mathcal{X} \in \mathcal{M}^N \mid \mathcal{F}(\mathcal{X}) \leq \mathcal{F}(\bar{\mathcal{X}})\} \quad (31)$$

for $\bar{\mathcal{X}} \in \mathcal{M}^N$, are compact.

Proof. First, observe from (15) that for every $\mathcal{X} \in \mathcal{M}^N$, the Riemannian exponential $\text{Exp}_{\mathcal{X}}$ is globally defined on $\mathcal{T}_{\mathcal{X}} \mathcal{M}^N$, which implies that $\mathcal{M}^N$ is geodesically complete.

Therefore, the Hopf-Rinow Theorem [33] implies that closed and bounded subsets of $\mathcal{M}^N$ are compact, and it suffices to show that sublevel sets of the form of $\mathcal{L}(\bar{\mathcal{X}})$ are closed and bounded.

From (10)-(13), $\mathcal{F}(\mathcal{X}) \geq 0$ for all $\mathcal{X} \in \mathcal{M}^N$, which, given continuity of $\mathcal{F}$ from Lemma 6, implies that the sublevel sets of $\mathcal{F}$ are closed. Turning to boundedness of sublevel sets, (1) implies that $\mathcal{M}$ is unbounded, and therefore $\mathcal{M}^N$ is unbounded. Then, by [34, Theorem 1], the sublevel set $\mathcal{L}(\bar{\mathcal{X}})$ is bounded if and only if $\mathcal{F}$ is coercive. Coercivity is defined in terms of the Riemannian distance $d_{\mathcal{M}^N}$ from (24) as follows: for all $\mathcal{Y} \in \mathcal{M}^N$, every sequence $\{\mathcal{X}_l\}_{l \in \mathbb{N}} \subset \mathcal{M}^N$ such that $\lim_{l \rightarrow \infty} d_{\mathcal{M}^N}(\mathcal{X}_l, \mathcal{Y}) = \infty$ also satisfies $\lim_{l \rightarrow \infty} \mathcal{F}(\mathcal{X}_l) = \infty$. Therefore, to complete the proof, it suffices to show that $\mathcal{F}$ is coercive, which we do next.

We consider a pose graph $\mathcal{V}$ that contains vertices whose corresponding poses are arbitrarily far apart. Let $\mathcal{X}_l \triangleq \text{vec}((\mathbf{x}_{lm}^p)_{(m,p) \in \mathcal{V}})$ and $\mathcal{Y} \triangleq \text{vec}((\mathbf{y}_m^p)_{(m,p) \in \mathcal{V}})$, and observe from the definition of $d_{\mathcal{M}^N}(\mathcal{X}_l, \mathcal{Y})$ in (24) that

$$\lim_{d_{\mathcal{M}^N}(\mathcal{X}_l, \mathcal{Y}) \rightarrow \infty} \max_{(m,p) \in \mathcal{V}} \|\text{Log}_{\mathbf{x}_{lm}^p}(\mathbf{y}_m^p)\|^2 = \infty.$$

Using $\mathbf{x}_{lm}^p \triangleq [\mathbf{x}_{lm,r}^p, \mathbf{x}_{lm,d}^p]^\top$ and $\mathbf{y}_m^p \triangleq [\mathbf{y}_{m,r}^p, \mathbf{y}_{m,d}^p]^\top$, it follows from (23) that

$$\begin{aligned}\|\text{Log}_{\mathbf{x}_{lm}^p}(\mathbf{y}_m^p)\|^2 &= \psi\left((\mathbf{x}_{lm,r}^p)^{-1} \otimes \mathbf{y}_{m,r}^p\right)^2 \\ &\quad + \|\mathbf{x}_{lm,d}^p - \mathbf{y}_{m,d}^p\|^2, \quad (32)\end{aligned}$$

where ψ is from Section IV-A. Applying the fact that $\psi(\cdot) \leq \pi$ and using the triangle inequality in (32) yields

$$\|\text{Log}_{\mathbf{x}_{lm}^p}(\mathbf{y}_m^p)\|^2 \leq \pi^2 + 2\|\mathbf{x}_{lm,d}^p\|^2 + 2\|\mathbf{y}_{m,d}^p\|^2,$$

which implies that

$$\lim_{\|\text{Log}_{\mathbf{x}_{lm}^p}(\mathbf{y}_m^p)\|^2 \rightarrow \infty} \max_{(m,p) \in \mathcal{V}} \|\mathbf{x}_{lm,d}^p\|^2 = \infty.$$

Furthermore, applying the definition of $\log(\cdot)$ in (4) gives

$$\|\log(\mathbf{x}_{lm}^p)\|^2 = \phi^2(\mathbf{x}_{lm}^p) + \frac{1}{\text{sinc}^2(\phi(\mathbf{x}_{lm}^p))} \|\mathbf{x}_{lm,d}^p\|^2, \quad (33)$$

where ϕ is from Section II-B. Applying $\text{sinc}^2(\cdot) \leq 1$ to (33) yields $\|\mathbf{x}_{lm,d}^p\|^2 \leq \|\log(\mathbf{x}_{lm}^p)\|^2$, and therefore $\lim_{\|\mathbf{x}_{lm,d}^p\|^2 \rightarrow \infty} \|\log(\mathbf{x}_{lm}^p)\|^2 = \infty$. For any $\mathbf{x}, \mathbf{y} \in \mathcal{M}$, we have

$$\begin{aligned}\|\log(\mathbf{x} \boxplus \mathbf{y})\|^2 &= \text{sinc}(\phi(\mathbf{x} \boxplus \mathbf{y}))^{-2} \mathbf{y}^\top M_L(\mathbf{x}) \mathbf{y} \\ &= \text{sinc}(\phi(\mathbf{x} \boxplus \mathbf{y}))^{-2} \mathbf{x}^\top M_R(\mathbf{y}) \mathbf{x},\end{aligned}$$

where $M_L(\mathbf{x}) \triangleq Q_L^\top(\mathbf{x}) \text{diag}(0, 1, 1, 1) Q_L(\mathbf{x})$ and $M_R(\mathbf{y}) \triangleq Q_R^\top(\mathbf{y}) \text{diag}(0, 1, 1, 1) Q_R(\mathbf{y})$. Because $M_L(\cdot), M_R(\cdot) \succeq 0$, it holds that, for any $\mathbf{x}, \mathbf{y} \in \mathcal{M}$,

$$\lim_{\|\log(\mathbf{x} \boxplus \mathbf{y})\|^2 \rightarrow \infty} \max \left\{ \|\log(\mathbf{x})\|^2, \|\log(\mathbf{y})\|^2 \right\} = \infty. \quad (34)$$

We now observe that weak connectivity of $\mathcal{G}$ from Assumption 1 implies that, for any two poses $\mathbf{x}_i^p, \mathbf{x}_j^q \in \mathcal{M}$ such that $(p, i), (q, j) \in \mathcal{V}$, there exists a sequence of

$L \geq 2$ weakly-connected vertices one can traverse from (p, i) to (q, j) . Renumbering this sequence yields vertices $(v_1, \dots, v_L) = ((p, i), \dots, (q, j))$ and associated edges $\mathcal{E}_L \subset \mathcal{E}$, poses $(\mathbf{x}_1, \dots, \mathbf{x}_L) = (\mathbf{x}_i^p, \dots, \mathbf{x}_j^q)$, and measurements $\{\tilde{\mathbf{z}}_{lm}\}_{(v_l, v_m) \in \mathcal{E}_L}$. Then, any two neighboring poses $\mathbf{x}_l, \mathbf{x}_{l+1}$ (such that $l, l+1 \in [L]$) have the property that $\mathbf{x}_{l+1} = \mathbf{x}_l \boxplus \mathbf{c}_{l(l+1)}$, where

$$\mathbf{c}_{l(l+1)} \triangleq \begin{cases} \tilde{\mathbf{z}}_{l(l+1)} \boxplus \mathbf{r}_{l(l+1)} & \text{if } (v_l, v_{l+1}) \in \mathcal{E}_L \\ \mathbf{r}_{l(l+1)}^{-1} \boxplus \tilde{\mathbf{z}}_{l(l+1)}^{-1} & \text{otherwise} \end{cases}, \quad (35)$$

with $\mathbf{r}_{l(l+1)} \triangleq \tilde{\mathbf{z}}_{l(l+1)}^{-1} \boxplus \mathbf{x}_l^{-1} \boxplus \mathbf{x}_{l+1}$. Therefore, we can inductively write $\mathbf{x}_L = \mathbf{x}_1 \boxplus \mathbf{C}_{1L}$, where $\mathbf{C}_{1L} \triangleq \mathbf{c}_{12} \boxplus \dots \boxplus \mathbf{c}_{(L-1)L}$.

Per Section III-C, we have fixed $\mathbf{x}_a^r = \mathbf{1}$ for some $(r, a) \in \mathcal{V}$, and applying the above composition to the sequence of vertices $(v_1, \dots, v_L) = ((r, a), \dots, (p, m))$ yields either $\mathbf{x}_{lm}^p = \mathbf{x}_a^r \boxplus \mathbf{C}_{1L}$ or $\mathbf{x}_{lm}^p = \mathbf{x}_a^r \boxplus \mathbf{C}_{1L}^{-1}$ depending on the ordering of vertices, with $\mathbf{C}_{1L}$ defined as above. However, since $\log(\mathbf{x}^{-1}) = -\log(\mathbf{x})$, it holds in both cases that $\|\log(\mathbf{x}_{lm}^p)\|^2 = \|\log(\mathbf{C}_{1L})\|^2$ for any $(p, m) \in \mathcal{V}$. Furthermore, because the $\tilde{\mathbf{z}}_{ij}$ terms in (35) are constant, applying (34) inductively yields, for any $(p, m) \in \mathcal{V}$,

$$\lim_{\|\log(\mathbf{x}_{lm}^p)\|^2 \rightarrow \infty} \max_{((b,i),(c,j)) \in \mathcal{E}} \|\log(\mathbf{r}_{ij}^{bc})\|^2 = \infty,$$

where $\mathbf{r}_{ij}^{bc} \triangleq (\tilde{\mathbf{z}}_{ij}^{bc})^{-1} \boxplus (\mathbf{x}_i^b)^{-1} \boxplus \mathbf{x}_j^c$. From (10)-(13), we have $\frac{1}{4} \lambda_{\min}(\Omega_{ij}^{bc}) \|\log(\mathbf{r}_{ij}^{bc})\|^2 \leq \mathcal{F}(\mathcal{X}_l)$, where $\lambda_{\min}(\cdot)$ is the minimum eigenvalue, and $\lambda_{\min}(\Omega_{ij}^{bc}) > 0$ because $\Omega_{ij}^{bc} = (\Sigma_{ij}^{bc})^{-1} \succ 0$. Therefore, $\lim_{\|\log(\mathbf{r}_{ij}^{bc})\|^2 \rightarrow \infty} \mathcal{F}(\mathcal{X}_l) = \infty$, and $\mathcal{F}$ is coercive. ■

We use Lemma 1 to show that the RGN Hessian approximation is bounded over any sublevel set.

Lemma 2 (Bounded RGN Hessian Approximation). *Suppose Assumption 1 holds. Fix $\tilde{\mathcal{X}} \in \mathcal{M}^N$, and let $\mathcal{L}(\tilde{\mathcal{X}})$ be the associated sublevel set of $\mathcal{F}$ from (31). Consider a point in $\mathcal{L}(\tilde{\mathcal{X}})$ of the form $\tilde{\mathcal{X}} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_R)$. For each p , define $(\tilde{X}_p, \tilde{X}_{S(p)}) \triangleq (\tilde{X}_p, \{\tilde{X}_q\}_{q \in S(p)})$. Then the matrix H_p defined in (28) is bounded over $\mathcal{L}(\tilde{\mathcal{X}})$ in the sense that there exists a constant $M > 0$ such that*

$$\|H_p(\tilde{X}_p, \tilde{X}_{S(p)})\| \leq M \quad (36)$$

for all $p \in \mathcal{R}$ and all points $(\tilde{X}_p, \tilde{X}_{S(p)})$ defined above.

Proof. Follows from Appendix J-C in [24]. ■

Next we show the Euclidean gradient of f_p is Lipschitz.

Lemma 3 (Euclidean Gradient is Lipschitz Continuous). *Fix any $\tilde{\mathcal{X}} \in \mathcal{M}^N$. Consider a point in the set $\mathcal{L}(\tilde{\mathcal{X}})$ from (31) of the form $\tilde{\mathcal{X}} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_R)$. For each p , define $(\tilde{X}_p, \tilde{X}_{S(p)}) \triangleq (\tilde{X}_p, \{\tilde{X}_q\}_{q \in S(p)})$. Define the set*

$$\mathcal{X}_p \triangleq \{\tilde{X}_p \in \mathcal{M}^{N_p} : \text{there exist points } Y_i \text{ for } i \in \mathcal{R} \setminus \{p\} \text{ such that } (Y_1, \dots, \tilde{X}_p, \dots, Y_R) \in \mathcal{L}(\tilde{\mathcal{X}})\}.$$

Then there exists some $L_E > 0$ such that the Euclidean gradient of the local cost f_p with respect to its first argument, denoted $\nabla_{X_p} f_p(\cdot, \tilde{X}_{S(p)})$, is L_E -Lipschitz continuous on $\mathcal{X}_p$.

Proof. A function's Euclidean gradient is Lipschitz continuous if its Euclidean Hessian is bounded. Results in [24, Appendix I] derive analytical formulas for all Hessian tensors of the cost $\mathcal{F}$ from Problem 1, and results in [24, Appendix K] show that their Frobenius norms are bounded on any compact set. Then the Euclidean gradient of $\mathcal{F}$ is Lipschitz continuous, and the result follows because the Euclidean gradient of f_p with respect to its first argument is Lipschitz with a constant no larger than that of the full Euclidean gradient of $\mathcal{F}$. ■

Similar to Lemma 3, we next establish that the Riemannian gradients of interest are Lipschitz.

Lemma 4 (Local Riemannian Gradient is Lipschitz Continuous). *Suppose Assumption 1 holds. Let $\varphi_p(X_p) \triangleq f_p(X_p, X_{S(p)})$ be the local cost function of robot p with all neighbor poses in $X_{S(p)}$ held constant. On any compact subset $\mathcal{K} \subset \mathcal{M}^{N_p}$, the Riemannian gradient $\text{grad } \varphi_p$ defined in (25) is L_g -Lipschitz continuous. That is, there exists a constant $L_g > 0$ such that for all $X_p, Y_p \in \mathcal{K}$ we have*

$$\|\mathcal{P}_{X_p \rightarrow Y_p} \text{grad } \varphi_p(X_p) - \text{grad } \varphi_p(Y_p)\| \leq L_g d_{\mathcal{M}^{N_p}}(X_p, Y_p),$$

where $\mathcal{P}_{X_p \rightarrow Y_p}$ denotes the parallel transport operator and $d_{\mathcal{M}^{N_p}}(\cdot, \cdot)$ is the product manifold distance from (24).

Proof. By [35, Lemma 2], $\text{grad } \varphi_p$ is L_g -Lipschitz if its Riemannian Hessian is bounded. The Riemannian Hessian of φ_p is bounded because $\text{Hess } \varphi_p$ is a principal sub-matrix of the global Riemannian Hessian $\text{Hess } \mathcal{F}$, which is proven to be bounded on compact sets in [24, Appendix G]. ■

The final technical lemma establishes a Lipschitz-like condition for the Riemannian exponential map we use in Algorithm 1.

Lemma 5. *For Exp_X in (15) there exists a constant $\delta > 0$ such that*

$$\|\text{Exp}_X(\mathbf{s}) - X\| \leq \delta \|\mathbf{s}\|$$

for any $X \in \mathcal{M}^N$ and $\mathbf{s} \in \mathcal{T}_X \mathcal{M}^N$.

Proof. Expanding the definition of Exp_X in (15) yields

$$\|\text{Exp}_X(\mathbf{s}) - X\|^2 = \sum_{i=1}^N \|\text{Exp}_{\mathbf{x}_i}(\mathbf{s}_i) - \mathbf{x}_i\|^2,$$

where $X = ((\mathbf{x}_i)_{i=1}^N)$, $\mathbf{x}_i = [\mathbf{x}_{i,r}^\top, \mathbf{x}_{i,d}^\top]^\top \in \mathcal{M}$, and $\mathbf{s}_i = [\mathbf{s}_{i,r}^\top, \mathbf{s}_{i,d}^\top]^\top \in \mathcal{T}_{\mathbf{x}_i} \mathcal{M}$. Substituting the definition of $\text{Exp}_{\mathbf{x}_i}$ from (14) gives

$$\begin{aligned} & \|\text{Exp}_X(\mathbf{s}) - X\|^2 \\ &= \sum_{i=1}^N \left(\|\mathbf{x}_{i,r} \otimes \psi^{-1} \left((\mathbf{x}_{i,r}^{-1} \otimes \mathbf{s}_{i,r})_2 \right) - \mathbf{x}_{i,r}\|^2 + \|\mathbf{s}_{i,d}\|^2 \right). \end{aligned} \quad (37)$$

Next, we observe that $\mathbf{x}_{i,r} \in S^1$ and $\mathbf{s}_{i,r} \in \mathcal{T}_{\mathbf{x}_{i,r}} S^1$, so that $\text{Exp}_{\mathbf{x}_{i,r}}(\mathbf{s}_{i,r}) \triangleq \mathbf{x}_{i,r} \otimes \psi^{-1} \left((\mathbf{x}_{i,r}^{-1} \otimes \mathbf{s}_{i,r})_2 \right)$ defines the Riemannian exponential map on S^1 at the point $\mathbf{x}_{i,r}$. The manifold S^1 is compact and smooth; therefore, by [36, Equation (31)], there exists a constant $\delta_r > 0$ such that

$$\|\mathbf{x}_{i,r} \otimes \psi^{-1} \left((\mathbf{x}_{i,r}^{-1} \otimes \mathbf{s}_{i,r})_2 \right) - \mathbf{x}_{i,r}\|^2 \leq \delta_r \|\mathbf{s}_{i,r}\|^2. \quad (38)$$

Substituting (38) into (37) yields

$$\begin{aligned} \|\text{Exp}_X(\mathbf{s}) - X\|^2 &\leq \sum_{i=1}^N \delta_r \|\mathbf{s}_{i,r}\|^2 + \|\mathbf{s}_{d_i}\|^2 \\ &\leq \sum_{i=1}^N \max\{\delta_r, 1\} (\|\mathbf{s}_{i,r}\|^2 + \|\mathbf{s}_{d_i}\|^2) \\ &= \sum_{i=1}^N \max\{\delta_r, 1\} \|\mathbf{s}_i\|^2 \\ &= \max\{\delta_r, 1\} \sum_{i=1}^N \|\mathbf{s}_i\|^2 \\ &= \max\{\delta_r, 1\} \|\mathbf{s}\|^2, \end{aligned}$$

where the last equality holds because the squared norm of a tangent vector on the product manifold is the sum of the squared norms of its components. Taking the square root of both sides completes the proof. ■

We next present an elementary lemma on the smoothness of the cost function $\mathcal{F}$.

Lemma 6 (Smoothness of Cost Function). *The function $\mathcal{F}$ from Problem 1 and the local cost functions f_p for $p \in \mathcal{R}$ are continuously differentiable.*

Proof. The functions $\mathcal{F}$ and f_p for $p \in \mathcal{R}$ are sums of the terms h_p from (11) and $s_{p \leftrightarrow q}$ from (12). These functions are sums of squared weighted norms of the tangent residual function e_{ij} from (9). The function e_{ij} was shown to be continuously differentiable in Appendix H of [24], and hence the result follows. ■

The next lemma relates the robots' gradients of their local costs to the descent directions they use in Algorithm 1.

Lemma 7 (Basic LM step bounds). *Fix $p \in \mathcal{R}$, and suppose robot p has $(X_p, X_{S(p)})$ onboard. Consider J_p from (30) and define*

$$g_p := \text{grad } f_p(X_p, X_{S(p)}), \quad H_p := J_p^\top J_p, \quad \text{and } \lambda > 0,$$

where f_p is from (10). Suppose s solves the Levenberg–Marquardt (LM) system

$$(H_p + \lambda I)s = g_p. \quad (39)$$

Then

$$\|g_p\|^2 \geq \lambda^2 \|s\|^2 \quad (40)$$

and

$$\langle g_p, s \rangle \geq \frac{\|g_p\|^2}{\|J_p\|^2 + \lambda}. \quad (41)$$

Proof. First, we prove (40). By (39),

$$\begin{aligned} \|g_p\|^2 &= \langle (H_p + \lambda I)s, (H_p + \lambda I)s \rangle \\ &= \|H_p s\|^2 + 2\lambda \langle H_p s, s \rangle + \lambda^2 \|s\|^2. \end{aligned} \quad (42)$$

From the expression in (28) we have $H_p = J_p^\top J_p \succeq 0$, and therefore

$$\langle H_p s, s \rangle = \langle J_p^\top J_p s, s \rangle = \langle J_p s, J_p s \rangle = \|J_p s\|^2 \geq 0.$$

Therefore all three terms on the right-hand side of (42) are nonnegative, and we may write $\|g_p\|^2 \geq \lambda^2 \|s\|^2$, which proves (40).

Next, we prove (41). Since $H_p := J_p^\top J_p \succeq 0$ and $\lambda I > 0$, we know that $H_p + \lambda I$ is positive definite and hence invertible. Then $s = (H_p + \lambda I)^{-1} g_p$ solves (39), which implies that

$$\langle g_p, s \rangle = \langle g_p, (H_p + \lambda I)^{-1} g_p \rangle \geq \frac{\|g_p\|^2}{\|H_p + \lambda I\|} = \frac{\|g_p\|^2}{\|J_p\|^2 + \lambda},$$

where we have used the eigenvalue bound for positive definite matrices from [37, Theorem 4.2.2] that shows $\langle u, A^{-1}u \rangle \geq \|u\|^2 / \|A\|$ for $A \succ 0$, and we have also used the relations $\|H_p + \lambda I\| = \|H_p\| + \lambda$ and $\|H_p\| = \|J_p^\top J_p\| = \|J_p\|^2$. ■

The following lemma establishes a certain Lipschitz-like condition for the robots' local costs.

Lemma 8 (Lipschitz-type Bound for the Pullback). *Suppose Assumption 1 holds. Fix $p \in \mathcal{R}$. Fix $\tilde{\mathcal{X}} \in \mathcal{M}^N$, and let $\mathcal{L}(\tilde{\mathcal{X}})$ be the associated sublevel set of $\mathcal{F}$ from (31). Consider a point $\tilde{\mathcal{X}} \in \mathcal{L}(\tilde{\mathcal{X}})$ of the form $\tilde{\mathcal{X}} = (\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_R)$. For each p , define $(\tilde{X}_p, \tilde{X}_{S(p)}) := (\tilde{X}_p, \{\tilde{X}_q\}_{q \in S(p)})$. Let $\varphi_p(\tilde{X}_p) \triangleq f_p(\tilde{X}_p, \tilde{X}_{S(p)})$ be robot p 's local cost with its neighbors' poses in $\tilde{X}_{S(p)}$ held constant. Suppose s solves (20) for a point of the form $(\tilde{X}_p, \tilde{X}_{S(p)})$, and define the pullback of φ_p at $\tilde{X}_p$ as*

$$\hat{\varphi}_p(s) \triangleq \varphi_p(\text{Exp}_{\tilde{X}_p}(s)) \quad \text{for } s \in T_{\tilde{X}_p} \mathcal{M}^{N_p},$$

where $\text{Exp}_{\tilde{X}_p}$ is from (15) and $T_{\tilde{X}_p} \mathcal{M}^{N_p}$ is from (2). Then there exists $\tilde{L}_g > 0$ such that

$$|\hat{\varphi}_p(-s) - [\varphi_p(\tilde{X}_p) + \langle \text{grad } \varphi_p(\tilde{X}_p), -s \rangle]| \leq \frac{\tilde{L}_g}{2} \|s\|^2. \quad (43)$$

Proof. The sublevel set $\mathcal{L}(\tilde{\mathcal{X}})$ is compact by Lemma 1. For the choice of $\lambda_k = 2\mu_{p_k} f_{p_k}(X_{p_k}, X_{S(p_k)})$ from (18), at any $(\tilde{X}_p, \tilde{X}_{S(p)})$, the bound in (40) gives

$$\|s\|^2 \leq \frac{\|\text{grad } \varphi_p(\tilde{X}_p)\|^2}{\mu_{\min}^2 \|F_p(\tilde{X}_p, \tilde{X}_{S(p)})\|^4}, \quad (44)$$

where F_p is from (29) and where we have used $\mu_{p_k} \geq \mu_{\min}$. The right-hand side of (44) is continuous in $\tilde{X}_p$ and non-negative because both F_p and $\text{grad } \varphi_p$ are continuous in $\tilde{X}_p$ and $\|\cdot\|$ is continuous and non-negative. The function φ_p is twice continuously differentiable by [24, Appendix I]. Then the result follows from Lemma 10.57 in [38]. ■

The next lemma establishes a Lipschitz-like condition for the robots' global cost.

Lemma 9 (Lipschitz-type Bound for $\mathcal{F}$). *Suppose Assumption 1 holds. Fix $\tilde{\mathcal{X}} \in \mathcal{M}^N$, and let $\mathcal{L}(\tilde{\mathcal{X}})$ be the associated sublevel set of $\mathcal{F}$ from (31). Consider a point $\tilde{\mathcal{X}} \in \mathcal{L}(\tilde{\mathcal{X}})$ of the form $\tilde{\mathcal{X}} = (\tilde{X}_1, \dots, \tilde{X}_R)$. Fix $p_k \in \mathcal{R}$, and suppose s_{p_k} solves (20) for this choice of p_k . Consider*

$$\hat{s} = (0, 0, \dots, s_{p_k}, \dots, 0, 0),$$

where s_{p_k} is the p_k^{th} entry of $\hat{s}$. Then, there exists $L_{\mathcal{F}} > 0$ such that

$$|\mathcal{F}(\text{Exp}_{\tilde{\mathcal{X}}}(-\hat{s})) - [\mathcal{F}(\tilde{\mathcal{X}}) + \langle \text{grad } \mathcal{F}(\tilde{\mathcal{X}}), -\hat{s} \rangle]| \leq \frac{L_{\mathcal{F}}}{2} \|\hat{s}\|^2,$$

where $\mathcal{F}$ is from (13).

Proof. The sublevel set $\mathcal{L}(\tilde{\mathcal{X}})$ is compact by Lemma 1. For the choice of $\lambda_k = 2\mu_{p_k} f_{p_k}(\tilde{X}_{p_k}, \tilde{X}_{S(p_k)})$ from (18), we can use (39) to find

$$\|\hat{s}\|^2 = \|s_{p_k}\|^2 \leq \max_{q \in \mathcal{R}} \frac{\|\text{grad } f_q(\tilde{X}_q, \tilde{X}_{S(q)})\|^2}{\mu_{\min}^2 \|F_q(\tilde{X}_q, \tilde{X}_{S(q)})\|^4}, \quad (45)$$

where f_q is from (10) and F_q is from (29), and where we have used $\mu_{p_k} \geq \mu_{\min}$. The right-hand side in (45) is continuous in $\tilde{\mathcal{X}}$ and non-negative because f_q and F_q are continuous in $\tilde{\mathcal{X}}$, the norm $\|\cdot\|$ is continuous and non-negative, and the maximum of continuous functions is also continuous. The function $\mathcal{F}$ is twice continuously differentiable by [24, Appendix I]. Then the result follows from Lemma 10.57 in [38]. ■

Appendix B gave expressions for the Riemannian gradient g_{p_k} and Riemannian Gauss-Newton (RGN) Hessian approximation H_{p_k} . In the next lemma, we use expressions for both of these objects that are equivalent but take a different form, namely

$$g_{p_k}(X_{p_k}(k), \hat{X}_{S(p_k)}) = J_{p_k}(D_{p_k})^\top F_{p_k}(D_{p_k}) \quad (46)$$

and

$$H_{p_k}(X_{p_k}, \hat{X}_{S(p_k)}) \triangleq J_{p_k}(D_{p_k})^\top J_{p_k}(D_{p_k}). \quad (47)$$

Lemma 10 (Predicted reduction upper bound). *Suppose robot p_k has $(X_{p_k}, \hat{X}_{S(p_k)})$ onboard at iteration k of Algorithm 1, and consider θ_{p_k} from (17). Then, for all $s \in T_{X_{p_k}} \mathcal{M}^{N_{p_k}}$,*

$$\begin{aligned} \frac{1}{2}(\theta_{p_k}(0) - \theta_{p_k}(-s)) &= \\ \langle g_{p_k}, s \rangle - \frac{1}{2}\|J_{p_k}s\|^2 - \frac{\lambda_k}{2}\|s\|^2 &\leq \langle g_{p_k}, s \rangle, \end{aligned}$$

where $F_{p_k} := F_{p_k}(X_{p_k}, \hat{X}_{S(p_k)})$ is from (29), $J_{p_k} := J_{p_k}(X_{p_k}, \hat{X}_{S(p_k)})$ is from (30), and $g_{p_k} := g_{p_k}(X_{p_k}, \hat{X}_{S(p_k)})$ is from (46).

Proof. Using (46) and (47) we have

$$\theta_{p_k}(0) = \|F_{p_k}\|^2 \quad \text{and} \quad \theta_{p_k}(-s) = \|F_{p_k} - J_{p_k}s\|^2 + \lambda_k\|s\|^2.$$

Hence

$$\begin{aligned} \frac{1}{2}(\theta_{p_k}(0) - \theta_{p_k}(-s)) &= \\ \frac{1}{2}(\|F_{p_k}\|^2 - \|F_{p_k} - J_{p_k}s\|^2 - \lambda_k\|s\|^2) &= \end{aligned} \quad (48)$$

Expand the square using the Euclidean inner product gives

$$\|F_{p_k} - J_{p_k}s\|^2 = \|F_{p_k}\|^2 - 2\langle F_{p_k}, J_{p_k}s \rangle + \|J_{p_k}s\|^2.$$

Substituting this expansion into (48) gives

$$\begin{aligned} \frac{1}{2}(\theta_{p_k}(0) - \theta_{p_k}(-s)) &= \\ \frac{1}{2}(\|F_{p_k}\|^2 - \|F_{p_k}\|^2 + 2\langle F_{p_k}, J_{p_k}s \rangle - \|J_{p_k}s\|^2 - \lambda_k\|s\|^2) &= \\ \langle F_{p_k}, J_{p_k}s \rangle - \frac{1}{2}\|J_{p_k}s\|^2 - \frac{\lambda_k}{2}\|s\|^2. \end{aligned}$$

We then apply the definition of the adjoint operator $J_{p_k}^\top$ to find

$$\langle F_{p_k}, J_{p_k}s \rangle = \langle J_{p_k}^\top F_{p_k}, s \rangle = \langle g_{p_k}, s \rangle.$$

Therefore

$$\frac{1}{2}(\theta_{p_k}(0) - \theta_{p_k}(-s)) = \langle g_{p_k}, s \rangle - \frac{1}{2}\|J_{p_k}s\|^2 - \frac{\lambda_k}{2}\|s\|^2.$$

Finally, since $\|J_{p_k}s\|^2 \geq 0$ and $\lambda_k\|s\|^2 \geq 0$, we obtain the upper bound

$$\frac{1}{2}(\theta_{p_k}(0) - \theta_{p_k}(-s)) \leq \langle g_{p_k}, s \rangle. \quad \blacksquare$$

Next we bound the reduction in a robot's cost that is incurred at an iteration of Algorithm 1.

Lemma 11 (Actual reduction lower bound via pullback). *Suppose Assumption 1 holds. Let φ_{p_k} and $\hat{\varphi}_{p_k}$ be the local cost function and its pullback to the tangent space, respectively, as defined in Lemma 8. The actual reduction in the local cost f_{p_k} achieved at line 10 of Algorithm 1 is defined as $\text{Actual}(-s) := \varphi_{p_k}(X_{p_k}) - \hat{\varphi}_{p_k}(-s)$. Suppose X_{p_k} and $\text{Exp}_{X_{p_k}}(-s)$ are both in the sublevel set $\mathcal{L}(\tilde{\mathcal{X}})$ for some point $\tilde{\mathcal{X}}$. Then, for any tangent vector $s \in T_{X_{p_k}} \mathcal{M}^{N_{p_k}}$, the actual reduction satisfies*

$$\text{Actual}(-s) \geq \langle g_{p_k}, s \rangle - \frac{L_g}{2}\|s\|^2,$$

where $g_{p_k} := g_{p_k}(X_{p_k}, \hat{X}_{S(p_k)})$ is the local Riemannian gradient from (46) and L_g is the Lipschitz constant of the Riemannian gradient on the compact sublevel set $\mathcal{L}(\tilde{\mathcal{X}})$ defined in Lemma 4.

Proof. We apply (43) to the tangent vector $-s$ to find

$$\left| \hat{\varphi}_{p_k}(-s) - [\varphi_{p_k}(X_{p_k}) + \langle g_{p_k}, -s \rangle] \right| \leq \frac{L_g}{2}\|s\|^2.$$

Using $\langle g_{p_k}, -s \rangle = -\langle g_{p_k}, s \rangle$, this bound becomes

$$\left| \hat{\varphi}_{p_k}(-s) - \varphi_{p_k}(X_{p_k}) + \langle g_{p_k}, s \rangle \right| \leq \frac{L_g}{2}\|s\|^2.$$

This absolute value inequality implies

$$\hat{\varphi}_{p_k}(-s) - \varphi_{p_k}(X_{p_k}) + \langle g_{p_k}, s \rangle \leq \frac{L_g}{2} \|s\|^2,$$

and therefore

$$\varphi_{p_k}(X_{p_k}) - \hat{\varphi}_{p_k}(-s) \geq \langle g_{p_k}, s \rangle - \frac{L_g}{2} \|s\|^2.$$

■

The following lemma bounds the Hessian approximation that the robots use in their computations.

Lemma 12 (Damped Hessian Operator Bounds). *Suppose Assumption 1 holds. Fix $\bar{\mathcal{X}} \in \mathcal{M}^N$, and let $\mathcal{L}(\bar{\mathcal{X}})$ be the associated sublevel set of $\mathcal{F}$ defined in (31). Let H_{p_k} denote the RGN Hessian approximation from (47). Suppose $f_{p_k}(X_{p_k}, \hat{X}_{S(p_k)}) \leq \mathcal{F}(\bar{\mathcal{X}})$ so that the point associated with $(X_{p_k}, \hat{X}_{S(p_k)})$ is in $\mathcal{L}(\bar{\mathcal{X}})$. Let $\lambda_k \geq 0$ be computed as in (18). Then for any tangent vector $s \in T_{X_{p_k}} \mathcal{M}^{N_{p_k}}$ we have $\|H_{p_k} + \lambda_k I\| \leq \zeta$ and $\langle s, (H_{p_k} + \lambda_k I)s \rangle \leq \zeta \|s\|^2$, where $\zeta := M + \beta\kappa$, M is from (36), β is from Algorithm 1, and κ is from Lemma 14.*

Proof. The operator norm of the damped Hessian $H_{p_k} + \lambda_k I$ can be bounded with the triangle inequality via

$$\|H_{p_k}(X_{p_k}, \hat{X}_{S(p_k)}) + \lambda_k I\| \leq \|H_{p_k}(X_{p_k}, \hat{X}_{S(p_k)})\| + \|\lambda_k I\|.$$

Lemma 2 shows that the RGN Hessian is bounded on $\mathcal{L}(\bar{\mathcal{X}})$ via $\|H_{p_k}\| \leq M$. At every iteration k , Algorithm 1 uses a damping parameter λ_k that satisfies $\lambda_k \leq \beta\kappa$. Then $\|H_{p_k} + \lambda_k I\| \leq M + \beta\kappa$, which gives the first bound. The second bound follows by observing that

$$\langle s, (H_{p_k} + \lambda_k I)s \rangle \leq \|H_{p_k} + \lambda_k I\| \|s\|^2.$$

■

Next, we relate a robot's gradient with delayed communications to what its gradient would be with delay-free communications.

Lemma 13 (Delayed Gradient difference bound). *Suppose Assumption 2 holds. Let $g_k \triangleq \text{grad}_{X_{p_k}} f_{p_k}(X_{p_k}(k), \hat{X}_{S(p_k)}(k))$ be the local gradient that is computed by robot p_k in line 3 at iteration k of Algorithm 1. Let $\bar{g}_k = \text{grad}_{X_{p_k}} f_{p_k}(X_{p_k}(k), X_{S(p_k)}(k))$ be what robot p_k would compute in line 3 at iteration k of Algorithm 1 if it had the most up-to-date shared pose estimates from each of its neighbors. Then*

$$\|\bar{g}_k - g_k\| \leq \frac{\Delta}{2} L_E \delta_{\max} \gamma_k \sum_{k'=k-B}^{k-1} \|s(k')\|,$$

where $\delta_{\max} > 0$ is a constant, $s(k') = (0, \dots, s_{p_{k'}}, \dots, 0)$ is the search direction used in line 10 of Algorithm 1 at iteration k' , and $p_{k'}$ is the index of the robot that performs an update at iteration k' .

Proof. We know from (10) that the local cost with up-to-date neighbor info $X_{S(p_k)}(k)$ is

$$\begin{aligned} f_{p_k}(X_{p_k}(k), X_{S(p_k)}(k)) \\ = h_{p_k}(X_{p_k}(k)) + \frac{1}{2} \sum_{q \in S(p_k)} s_{p_k \leftrightarrow q}(X_{p_k}(k), X_q(k)), \end{aligned}$$

and the local cost using outdated neighbor info $\hat{X}_{S(p_k)}(k)$ is

$$\begin{aligned} f_{p_k}(X_{p_k}(k), \hat{X}_{S(p_k)}(k)) \\ = h_{p_k}(X_{p_k}(k)) + \frac{1}{2} \sum_{q \in S(p_k)} s_{p_k \leftrightarrow q}(X_{p_k}(k), \hat{X}_q^p(k)). \end{aligned}$$

Here $\hat{X}_q^p(k) = X_q(\ell_q^p(k))$ is the outdated trajectory that robot p has for neighbor $q \in S(p_k)$ at iteration k . The Riemannian gradient of f_{p_k} with respect to its first argument evaluated with up-to-date information is

$$\begin{aligned} \bar{g}_k &= \text{grad}_{X_{p_k}} f_{p_k}(X_{p_k}(k), X_{S(p_k)}(k)) \\ &= \text{grad}_{X_{p_k}} h_{p_k}(X_{p_k}(k)) \\ &\quad + \frac{1}{2} \sum_{q \in S(p_k)} \text{grad}_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), X_q(k)), \end{aligned} \quad (49)$$

and the Riemannian gradient computed by robot p_k with outdated information is

$$\begin{aligned} g_k &= \text{grad}_{X_{p_k}} f_{p_k}(X_{p_k}(k), \hat{X}_{S(p_k)}(k)) \\ &= \text{grad}_{X_{p_k}} h_{p_k}(X_{p_k}(k)) \\ &\quad + \frac{1}{2} \sum_{q \in S(p_k)} \text{grad}_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), \hat{X}_q^p(k)). \end{aligned} \quad (50)$$

Using (49) and (50) to compute $\|g_k - \bar{g}_k\|$ gives

$$\begin{aligned} \|\bar{g}_k - g_k\| &= \left\| \frac{1}{2} \sum_{q \in S(p_k)} \left[\text{grad}_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), X_q(k)) \right. \right. \\ &\quad \left. \left. - \text{grad}_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), \hat{X}_q^p(k)) \right] \right\| \\ &\leq \frac{1}{2} \sum_{q \in S(p_k)} \left\| \text{grad}_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), X_q(k)) \right. \\ &\quad \left. - \text{grad}_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), \hat{X}_q^p(k)) \right\|. \end{aligned} \quad (51)$$

To bound each term, we use the fact that the Riemannian gradient with respect to X_{p_k} is the orthogonal projection of the Euclidean gradient onto the tangent space $T_{X_{p_k}(k)} \mathcal{M}^{N_p}$ (cf. Appendix B). That is, for all $q \in S(p_k)$,

$$\begin{aligned} \text{grad}_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), X_q(k)) \\ = \mathcal{P}_{X_{p_k}(k)} \nabla_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), X_q(k)) \end{aligned} \quad (52)$$

and

$$\begin{aligned} \text{grad}_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), \hat{X}_q^p(k)) \\ = \mathcal{P}_{X_{p_k}(k)} \nabla_{X_{p_k}} s_{p_k \leftrightarrow q}(X_{p_k}(k), \hat{X}_q^p(k)), \end{aligned} \quad (53)$$

where $\mathcal{P}_{X_{p_k}(k)}$ is from (26), and where $\nabla_{X_{p_k}} s_{p_k \leftrightarrow q}$ denotes the Euclidean gradient of $s_{p_k \leftrightarrow q}$ with respect to its first argument.

We apply (52) and (53) on the right hand side of (51) to reach

$$\begin{aligned} & \left\| \text{grad}_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), X_q(k)) \right. \\ & \quad \left. - \text{grad}_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), \hat{X}_q^p(k)) \right\| \\ &= \left\| \mathcal{P}_{X_{p_k}(k)} \nabla_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), X_q(k)) \right. \\ & \quad \left. - \mathcal{P}_{X_{p_k}(k)} \nabla_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), \hat{X}_q^p(k)) \right\| \end{aligned}$$

for all $q \in S(p_k)$.

Since the orthogonal projection operator $\mathcal{P}_{X_{p_k}(k)}$ is a non-expansive mapping, we have

$$\begin{aligned} & \left\| \mathcal{P}_{X_{p_k}(k)} \nabla_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), X_q(k)) \right. \\ & \quad \left. - \mathcal{P}_{X_{p_k}(k)} \nabla_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), \hat{X}_q^p(k)) \right\| \\ & \leq \left\| \nabla_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), X_q(k)) \right. \\ & \quad \left. - \nabla_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), \hat{X}_q^p(k)) \right\| \end{aligned}$$

for all $q \in S(p_k)$.

From Lemma 3 the Euclidean gradient of the local cost f_{p_k} is L_E -Lipschitz. Since f_{p_k} is the sum of h_{p_k} and $s_{p_k \leftrightarrow q}$, each Euclidean gradient term of the form $\nabla_{X_{p_k} s_{p_k \leftrightarrow q}}$ must also be Lipschitz continuous with a constant no greater than L_E . Then,

$$\begin{aligned} & \left\| \text{grad}_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), X_q(k)) \right. \\ & \quad \left. - \text{grad}_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), \hat{X}_q^p(k)) \right\| \\ & \leq \left\| \nabla_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), X_q(k)) \right. \\ & \quad \left. - \nabla_{X_{p_k} s_{p_k \leftrightarrow q}}(X_{p_k}(k), \hat{X}_q^p(k)) \right\| \\ & \leq L_E \|X_q(k) - \hat{X}_q^p(k)\| \end{aligned} \quad (54)$$

for all $q \in S(p_k)$.

Now we bound $\|X_q(k) - \hat{X}_q^p(k)\| = \|X_q(k) - X_q(\ell_q^p(k))\|$. By definition we have

$$X_q(k) - X_q(\ell_q^p(k)) = \sum_{k'=\ell_q^p(k)}^{k-1} X_q(k'+1) - X_q(k')$$

for all $q \in S(p_k)$, and taking the norm of both sides and using the triangle inequality gives

$$\|X_q(k) - X_q(\ell_q^p(k))\| \leq \sum_{k'=\ell_q^p(k)}^{k-1} \|X_q(k'+1) - X_q(k')\|. \quad (55)$$

Recall that for all $q \in S(p_k)$ and iterations k' , the next iterate $X_q(k'+1)$ is obtained from $X_q(k')$ via the update rule in line 10 of Algorithm 1. The value of X_q only changes if robot q was selected to perform an update at iteration k' , i.e., if $p_{k'} = q$, in which case

$$X_q(k'+1) = \text{Exp}_{X_q(k')}(-\gamma_k(s(k'))_q),$$

where $(s(k'))_q$ denotes the search direction $s(k')$ corresponding to robot q 's decision variables. If $p_{k'} \neq q$, then $(s(k'))_q = 0$ and $X_q(k'+1) = X_q(k')$.

From Lemma 5, we have

$$\|\text{Exp}_{X_q}(s_q) - X_q\| \leq \delta_q \|s_q\|$$

for all $X_q \in \mathcal{M}^{N_q}$, all $s_q \in T_{X_q} \mathcal{M}^{N_q}$, and some $\delta_q > 0$.

Then we bound the right-hand side of (55) as

$$\begin{aligned} & \|X_q(k) - X_q(\ell_q^p(k))\| \\ & \leq \sum_{k'=\ell_q^p(k)}^{k-1} \|\text{Exp}_{X_q(k')}(-\gamma_k(s(k'))_q) - X_q(k')\| \\ & \leq \sum_{k'=\ell_q^p(k)}^{k-1} \delta_q \|-\gamma_k(s(k'))_q\|, \end{aligned}$$

which holds for all $q \in S(p_k)$.

From Assumption 2 we have $\ell_q^p(k) \geq k - B$, which gives

$$\|X_q(k) - X_q(\ell_q^p(k))\| \leq \delta_q \sum_{k'=k-B}^{k-1} \|\gamma_k(s(k'))_q\|. \quad (56)$$

Then we substitute (56) into (54), and computing the sum over $q \in S(p_k)$ from (51) gives

$$\begin{aligned} \|\bar{g}_k - g_k\| & \leq \frac{1}{2} L_E \delta_q \gamma_k \sum_{q \in S(p_k)} \sum_{k'=k-B}^{k-1} \|(s(k'))_q\| \\ & = \frac{1}{2} L_E \delta_q \gamma_k \sum_{k'=k-B}^{k-1} \sum_{q \in S(p_k)} \|(s(k'))_q\| \\ & = \frac{1}{2} L_E \delta_q \gamma_k \sum_{k'=k-B}^{k-1} \deg(q) \|(s(k'))_q\| \\ & \leq \frac{1}{2} L_E \delta_q \gamma_k \sum_{k'=k-B}^{k-1} \Delta \|(s(k'))_q\|, \end{aligned} \quad (57)$$

where $\deg(q)$ is the degree of robot q in the given pose graph and where we have used $\deg(q) \leq \Delta$, where Δ is from (6).

Since $(s(k'))_q = 0$ unless $q = p_{k'}$, we simplify (57) to

$$\begin{aligned} \|\bar{g}_k - g_k\| & \leq \frac{1}{2} L_E \delta_q \gamma_k \sum_{k'=k-B}^{k-1} \Delta \|(s(k'))_q\| \\ & = \frac{\Delta}{2} L_E \delta_q \gamma_k \sum_{k'=k-B}^{k-1} \|s(k')\|, \end{aligned}$$

and setting $\delta_{\max} = \max_{q \in \mathcal{R}} \delta_q$ completes the proof. $\blacksquare$

E. Proof of Theorem 1.

Before proving Theorem 1, we state the following lemma.

Lemma 14 (Sufficient condition for acceptance). *Let $\eta \in (0, 1)$. Suppose robot $p_k \in \mathcal{R}$ is selected to compute an update at iteration k of Algorithm 1. If λ_k satisfies*

$$(1 - \eta) \lambda_k^2 \gamma_k - \frac{L_g}{2} \lambda_k - \frac{L_g}{2} M \geq 0, \quad (58)$$

where M is from Lemma 2, then the RLM acceptance ratio ρ_k from (22) satisfies $\rho_k \geq \eta$. In particular, it suffices to have $\lambda_k \geq \kappa$, where

$$\kappa \triangleq \frac{1}{2(1-\eta)\gamma_{\min}} \left(\frac{L_g}{2} + \sqrt{\frac{L_g^2}{4} + 2(1-\eta)L_g M} \right)$$

and $\gamma_{\min} = \min_{k \in \{0,1,\dots,K\}} \gamma_k$.

Proof. Consider φ_p from Lemma 8. Then the acceptance ratio defined in (22) can be written as

$$\begin{aligned} \rho_k &= \frac{\text{Actual}(-\gamma_k s_{p_k})}{\text{Predicted}(-\gamma_k s_{p_k})} \\ &= \frac{\varphi_{p_k}(X_{p_k}) - \varphi_{p_k}(\text{Exp}_{X_{p_k}}(-\gamma_k s_{p_k}))}{\frac{1}{2}(\theta_{p_k}(0) - \theta_{p_k}(-\gamma_k s_{p_k}))}, \end{aligned}$$

where $\hat{X}_{S(p_k)}$ is held constant. The condition $\rho_k \geq \eta$ is equivalent to having

$$\text{Actual}(-\gamma_k s_{p_k}) - \eta \cdot \text{Predicted}(-\gamma_k s_{p_k}) \geq 0.$$

From Lemma 11, we have a lower bound on the actual reduction in φ_{p_k} at iteration k , namely

$$\text{Actual}(-\gamma_k s_{p_k}) \geq \langle g_{p_k}, \gamma_k s_{p_k} \rangle - \frac{L_g}{2} \|\gamma_k s_{p_k}\|^2, \quad (59)$$

where $g_{p_k} := g_{p_k}(X_{p_k}, \hat{X}_{S(p_k)})$.

From Lemma 10, we have an upper bound on the predicted reduction in the form of

$$\text{Predicted}(-\gamma_k s_{p_k}) \leq \langle g_{p_k}, \gamma_k s_{p_k} \rangle,$$

and multiplying by $-\eta$ gives

$$-\eta \cdot \text{Predicted}(-\gamma_k s_{p_k}) \geq -\eta \langle g_{p_k}, \gamma_k s_{p_k} \rangle. \quad (60)$$

Adding (59) and (60), we find

$$\begin{aligned} &\text{Actual}(-\gamma_k s_{p_k}) - \eta \cdot \text{Predicted}(-\gamma_k s_{p_k}) \\ &\geq \left(\langle g_{p_k}, \gamma_k s_{p_k} \rangle - \frac{L_g}{2} \|\gamma_k s_{p_k}\|^2 \right) - \eta \langle g_{p_k}, \gamma_k s_{p_k} \rangle \\ &= (1-\eta) \langle g_{p_k}, \gamma_k s_{p_k} \rangle - \frac{L_g}{2} \|\gamma_k s_{p_k}\|^2. \quad (61) \end{aligned}$$

Next, we use Lemma 7 to write

$$\langle g_{p_k}, \gamma_k s_{p_k} \rangle \geq \gamma_k \frac{\|g_{p_k}\|^2}{\|J_{p_k}\|^2 + \lambda_k}, \quad (62)$$

where $J_{p_k} \triangleq J_{p_k}(X_{p_k}(k), \hat{X}_{S(p_k)}(k))$ is the local Riemannian Jacobian defined in (30). We can also write

$$\gamma_k^2 \|s_{p_k}\|^2 \leq \frac{\|g_{p_k}\|^2}{\lambda_k^2}, \quad (63)$$

which follows from (40).

We substitute (62) and (63) into (61) to find

$$\begin{aligned} &(1-\eta) \langle g_{p_k}, \gamma_k s_{p_k} \rangle - \frac{L_g}{2} \|\gamma_k s_{p_k}\|^2 \\ &\geq (1-\eta) \gamma_k \left(\frac{\|g_{p_k}\|^2}{\|J_{p_k}\|^2 + \lambda_k} \right) - \frac{L_g}{2} \left(\frac{\|g_{p_k}\|^2}{\lambda_k^2} \right). \quad (64) \end{aligned}$$

Using Lemma 2 and (47), we have $\|J_{p_k}\|^2 \leq M$. Then

$$\frac{1}{\|J_{p_k}\|^2 + \lambda_k} \geq \frac{1}{M + \lambda_k}. \quad (65)$$

Substituting (65) into (64) gives

$$\begin{aligned} &(1-\eta) \gamma_k \left(\frac{\|g_{p_k}\|^2}{\|J_{p_k}\|^2 + \lambda_k} \right) - \frac{L_g}{2} \left(\frac{\|g_{p_k}\|^2}{\lambda_k^2} \right) \\ &\geq (1-\eta) \gamma_k \left(\frac{\|g_{p_k}\|^2}{M + \lambda_k} \right) - \frac{L_g}{2} \left(\frac{\|g_{p_k}\|^2}{\lambda_k^2} \right) \\ &= \|g_{p_k}\|^2 \left[\frac{\gamma_k(1-\eta)}{M + \lambda_k} - \frac{L_g}{2\lambda_k^2} \right]. \end{aligned}$$

The step is successful, i.e., $\rho_k \geq \eta$, if this entire expression is non-negative. Since $\|g_{p_k}\|^2 \geq 0$, we only need to ensure that

$$\frac{\gamma_k(1-\eta)}{M + \lambda_k} - \frac{L_g}{2\lambda_k^2} \geq 0,$$

which holds if and only if

$$\frac{\gamma_k(1-\eta)}{M + \lambda_k} \geq \frac{L_g}{2\lambda_k^2}.$$

Solving for λ_k we recover the condition in (58).

Now we need to show that if $\lambda_k \geq \kappa$, then the inequality in (58) holds. Let $P(\lambda_k) = (1-\eta)\lambda_k^2\gamma_k - \frac{L_g}{2}\lambda_k - \frac{L_g}{2}M$. Since $\eta \in (0, 1)$, the coefficient of λ_k^2 is positive, and $P(\lambda_k)$ is an upward-opening parabola. Then, $P(\lambda_k) \geq 0$ for all λ_k greater than or equal to the largest root of $P(\lambda_k)$. The roots of $P(\lambda_k) = 0$ are

$$\lambda_k = \frac{\frac{L_g}{2} \pm \sqrt{\frac{L_g^2}{4} + 2(1-\eta)\gamma_k L_g M}}{2(1-\eta)\gamma_k}.$$

Since all terms under the square root are positive, the discriminant is real and positive. The largest root is

$$\frac{\frac{L_g}{2} + \sqrt{\frac{L_g^2}{4} + 2(1-\eta)\gamma_k L_g M}}{2(1-\eta)\gamma_k}.$$

Therefore, if $\lambda_k \geq \kappa$, then the condition (58) is satisfied. $\blacksquare$

We now prove Theorem 1.

Proof of Theorem 1: Line 6 of Algorithm 1 computes the initial value of λ_k as $\lambda_k = 2\mu_0 f_{p_k}(D_{p_k})$. Whenever a step is rejected, the algorithm updates $\mu_k \leftarrow \beta \mu_k$ with $\beta > 1$. After $j > 0$ unsuccessful updates, the damping parameter becomes $\lambda_k = 2(\mu_0 \beta^j) f_{p_k}(D_{p_k})$. From Lemma 14, we know that a step is guaranteed to succeed once $\lambda_k \geq \kappa$. The result follows by using $\mu_0 \geq \mu_{\min}$ and solving for the smallest integer j that satisfies $2\mu_0 \beta^j f_{p_k}(D_{p_k}) \geq \kappa$.

F. Decreasing Cost Lemma

The inner damping loop of Algorithm 1 (Lines 5-13) ensures a decrease in the local cost $f_{p_k}(\cdot, \hat{X}_{S(p_k)})$ with outdated information by enforcing the condition $\rho_k \geq \eta$. The next lemma shows that, under a certain step size bound, this decrease using outdated information also guarantees a

decrease in the global cost function $\mathcal{F}$ when evaluated with up-to-date information.

Lemma 15 (Cumulative Descent Property with Delays). *Suppose Assumptions 1 and 2 hold. Define $\mathcal{X}(k) \triangleq (X_1(k), X_2(k), \dots, X_R(k)) \in \mathcal{M}^N$ as the collection of all robots' estimates of their own trajectories at iteration k . If the step size $\gamma_k \in (0, 1]$ is chosen such that*

$$0 < \gamma_k < \frac{2\lambda_k}{(L_g + \Delta L_E \delta_{\max} B)}, \quad (66)$$

then the total cost reduction over K iterations satisfies

$$\mathcal{F}(\mathcal{X}(0)) - \mathcal{F}(\mathcal{X}(K)) \geq \sum_{k=0}^{K-1} \sigma_k \|s_{p_k}\|^2, \quad (67)$$

where $\sigma_k = \gamma_k \lambda_k - \gamma_k^2 \left(\frac{L_g}{2} + \frac{\Delta L_E \delta_{\max} B}{2} \right) > 0$, λ_k is from line 6 of Algorithm 1, L_g is from Lemma 4, Δ is from (6), L_E is from Lemma 3, $\delta_{\max} > 0$ is a constant, and B is from Assumption 2.

Proof. Suppose that at iteration k robot p_k is selected to compute an update. Recall that $\mathcal{X}(k) = (X_1(k), \dots, X_R(k)) \in \mathcal{M}^N$. For all $k \in \mathbb{N} \cup \{0\}$, define the aggregate tangent vector $s(k) \in T_{\mathcal{X}(k)} \mathcal{M}^N$ that represents the combined update directions for all robots as

$$s(k) = (0, 0, \dots, \underbrace{s_{p_k}}_{\text{position } p_k}, \dots, 0, 0),$$

where s_{p_k} is the local search direction computed via (20). That is, the p^{th} component of $s(k)$ is

$$(s(k))_p = \begin{cases} s_{p_k} & \text{if } p = p_k \\ 0 & \text{if } p \neq p_k \end{cases}. \quad (68)$$

We take $s(k) = 0$ for $k < 0$. By definition of $s(k)$, the update step in line 10 of Algorithm 1 is equivalent to

$$\mathcal{X}(k+1) = \text{Exp}_{\mathcal{X}(k)}(-\gamma_k s(k)).$$

By Lemma 9, we have

$$\begin{aligned} |\mathcal{F}(\mathcal{X}(k+1)) - [\mathcal{F}(\mathcal{X}(k)) + \langle \text{grad } \mathcal{F}(\mathcal{X}(k)), -\gamma_k s(k) \rangle]| \\ \leq \frac{L_g}{2} \|s(k)\|^2 \gamma_k^2, \end{aligned}$$

which implies that

$$\begin{aligned} -\frac{L_g}{2} \|s(k)\|^2 \gamma_k^2 &\leq \mathcal{F}(\mathcal{X}(k+1)) \\ &- [\mathcal{F}(\mathcal{X}(k)) + \langle \text{grad } \mathcal{F}(\mathcal{X}(k)), -\gamma_k s(k) \rangle] \\ &\leq \frac{L_g}{2} \|s(k)\|^2 \gamma_k^2. \end{aligned} \quad (69)$$

Using the right-hand inequality of (69) we find

$$\begin{aligned} \mathcal{F}(\mathcal{X}(k+1)) - \mathcal{F}(\mathcal{X}(k)) \\ \leq -\langle \text{grad } \mathcal{F}(\mathcal{X}(k)), \gamma_k s(k) \rangle + \frac{L_g}{2} \|s(k)\|^2 \gamma_k^2. \end{aligned}$$

From (68), we see that $s(k)$ is zero in all components $p \neq p_k$. Thus,

$$\begin{aligned} \mathcal{F}(\mathcal{X}(k+1)) - \mathcal{F}(\mathcal{X}(k)) \\ \leq -\langle \text{grad } \mathcal{F}(\mathcal{X}(k)), \gamma_k s(k) \rangle + \frac{L_g}{2} \|s(k)\|^2 \gamma_k^2 \\ = -\langle \text{grad}_{X_{p_k}} \mathcal{F}(\mathcal{X}(k)), \gamma_k (s(k))_{p_k} \rangle + \frac{L_g}{2} \|s_{p_k}\|^2 \gamma_k^2. \end{aligned} \quad (70)$$

Now, observe that

$$\begin{aligned} \text{grad}_{X_{p_k}} \mathcal{F}(\mathcal{X}) &= \sum_{p=1}^R \text{grad}_{X_{p_k}} h_p(X_p) \\ &+ \frac{1}{4} \text{grad}_{X_{p_k}} \left(\sum_{q \in S(p)} s_{p \leftrightarrow q}(X_p, X_q) \right. \\ &\quad \left. + \sum_{\{q \neq p | p \in S(q)\}} s_{q \leftrightarrow p}(X_q, X_p) \right). \end{aligned}$$

Since $q \in S(p)$ implies that $p \in S(q)$, and $s_{p \leftrightarrow q}(X_p, X_q) = s_{q \leftrightarrow p}(X_q, X_p)$, we have

$$\begin{aligned} \text{grad}_{X_{p_k}} \mathcal{F}(\mathcal{X}) &= \sum_{p=1}^R \text{grad}_{X_{p_k}} h_p(X_p) \\ &+ \frac{1}{2} \left(\sum_{q \in S(p)} \text{grad}_{X_{p_k}} s_{p \leftrightarrow q}(X_p, X_q) \right) \\ &= \text{grad}_{X_{p_k}} f_p(X_p, X_{S(p)}). \end{aligned}$$

Applying this to (70) yields

$$\mathcal{F}(\mathcal{X}(k+1)) - \mathcal{F}(\mathcal{X}(k)) \leq -\langle \bar{g}_k, \gamma_k s_{p_k} \rangle + \frac{L_g}{2} \|s_{p_k}\|^2 \gamma_k^2, \quad (71)$$

where $\bar{g}_k = \text{grad}_{X_{p_k}} f_{p_k}(X_{p_k}(k), X_{S(p_k)}(k))$ is the local gradient that would be computed by robot p_k at iteration k with all robots' up-to-date pose estimates. We let $g_k \triangleq \text{grad}_{X_{p_k}} f_{p_k}(X_{p_k}(k), \hat{X}_{S(p_k)}(k))$ be the outdated local gradient that robot p_k computes at iteration k .

The first term on the right-hand side of (71) can be expressed as

$$\begin{aligned} \langle \bar{g}_k, \gamma_k s_{p_k} \rangle &= \langle g_k + (\bar{g}_k - g_k), \gamma_k s_{p_k} \rangle \\ &= \langle g_k, \gamma_k s_{p_k} \rangle + \langle (\bar{g}_k - g_k), \gamma_k s_{p_k} \rangle. \end{aligned} \quad (72)$$

For the first inner product in (72) we use (20) to find

$$\begin{aligned} \langle g_k, \gamma_k s_{p_k} \rangle &= \langle (J_k^\top J_k + \lambda_k I) s_{p_k}, \gamma_k s_{p_k} \rangle \\ &= \|J_k s_{p_k}\|^2 \gamma_k + \lambda_k \|s_{p_k}\|^2 \gamma_k \geq \lambda_k \|s_{p_k}\|^2 \gamma_k, \end{aligned} \quad (73)$$

where $J_k \triangleq J_{p_k}(X_{p_k}(k), \hat{X}_{S(p_k)}(k))$ is from (30). For the second inner product in (72), we have

$$\begin{aligned} -\gamma_k \|\bar{g}_k - g_k\| \|s_{p_k}\| &\leq \langle (\bar{g}_k - g_k), \gamma_k s_{p_k} \rangle \\ &\leq \gamma_k \|\bar{g}_k - g_k\| \|s_{p_k}\|. \end{aligned} \quad (74)$$

We take the left inequality in (74) and combine it with the result of Lemma 13 to reach

$$\begin{aligned} \langle (\bar{g}_k - g_k), \gamma_k s_{p_k} \rangle &\geq -\gamma_k \|\bar{g}_k - g_k\| \|s_{p_k}\| \\ &\geq -\frac{\Delta}{2} L_E \delta_{\max} \gamma_k \left(\sum_{k'=k-B}^{k-1} \|s(k')\| \right) \|s_{p_k}\| \gamma_k = -A_k \gamma_k^2, \end{aligned} \quad (75)$$

where $A_k = \frac{\Delta L_E \delta_{\max}}{2} \|s_{p_k}\| \left(\sum_{k'=k-B}^{k-1} \|s(k')\| \right) > 0$.

Now we substitute (73) and (75) into (72) to find

$$\langle \bar{g}_k, \gamma_k s_{p_k} \rangle \geq \lambda_k \|s_{p_k}\|^2 \gamma_k - A_k \gamma_k^2. \quad (76)$$

We plug (76) into (70) to obtain

$$\begin{aligned} \mathcal{F}(\mathcal{X}(k+1)) - \mathcal{F}(\mathcal{X}(k)) &\leq -\langle \bar{g}_k, \gamma_k s_{p_k} \rangle + \frac{L_g}{2} \|s_{p_k}\|^2 \gamma_k^2 \\ &\leq -(\lambda_k \|s_{p_k}\|^2 \gamma_k - A_k \gamma_k^2) + \frac{L_g}{2} \|s_{p_k}\|^2 \gamma_k^2 \\ &= -\lambda_k \|s_{p_k}\|^2 \gamma_k + A_k \gamma_k^2 + \frac{L_g}{2} \|s_{p_k}\|^2 \gamma_k^2. \end{aligned} \quad (77)$$

To prove convergence, we analyze the total reduction in the cost function over K iterations. Summing (77) from $k = 0$ to $K - 1$, the left-hand side forms a telescoping sum of the form

$$\begin{aligned} \mathcal{F}(\mathcal{X}(K)) - \mathcal{F}(\mathcal{X}(0)) &= \sum_{k=0}^{K-1} (\mathcal{F}(\mathcal{X}(k+1)) - \mathcal{F}(\mathcal{X}(k))) \\ &\leq \sum_{k=0}^{K-1} \left[\left(\frac{L_g \gamma_k^2}{2} - \lambda_k \gamma_k \right) \|s_{p_k}\|^2 \right. \\ &\quad \left. + \frac{\Delta L_E \delta_{\max} \gamma_k^2}{2} \|s_{p_k}\| \sum_{k'=k-B}^{k-1} \|s(k')\| \right]. \end{aligned}$$

Rearranging to isolate the total cost reduction gives

$$\begin{aligned} \mathcal{F}(\mathcal{X}(0)) - \mathcal{F}(\mathcal{X}(K)) &\geq \sum_{k=0}^{K-1} \left(\left(\gamma_k \lambda_k - \gamma_k^2 \frac{L_g}{2} \right) \|s_{p_k}\|^2 \right. \\ &\quad \left. - \frac{\Delta L_E \delta_{\max}}{2} \gamma_k^2 \sum_{k'=k-B}^{k-1} \|s_{p_k}\| \cdot \|s(k')\| \right). \end{aligned} \quad (78)$$

We next use $\|s_{p_k}\| \|s(k')\| \leq \frac{1}{2} (\|s_{p_k}\|^2 + \|s(k')\|^2)$ to find

$$\begin{aligned} \sum_{k=0}^{K-1} \sum_{k'=k-B}^{k-1} \|s_{p_k}\| \|s(k')\| &\leq \sum_{k=0}^{K-1} \sum_{k'=k-B}^{k-1} \frac{1}{2} (\|s_{p_k}\|^2 + \|s(k')\|^2) \\ &= \frac{1}{2} \sum_{k=0}^{K-1} \sum_{k'=k-B}^{k-1} \|s_{p_k}\|^2 + \frac{1}{2} \sum_{k=0}^{K-1} \sum_{k'=k-B}^{k-1} \|s(k')\|^2. \end{aligned} \quad (79)$$

We bound these sums explicitly using the maximum delay B . For the first term on the right-hand side of (79), since the

summand $\|s_{p_k}\|^2$ is independent of the inner index k' , and the inner sum has at most B non-zero terms, we have

$$\frac{1}{2} \sum_{k=0}^{K-1} \sum_{k'=k-B}^{k-1} \|s_{p_k}\|^2 \leq \frac{1}{2} \sum_{k=0}^{K-1} B \|s_{p_k}\|^2 = \frac{B}{2} \sum_{k=0}^{K-1} \|s_{p_k}\|^2. \quad (80)$$

For the second term on the right-hand side of (79), we observe that any specific iteration index j appears in the inner sum only when $k - B \leq j \leq k - 1$. Thus, each term of the form $\|s_j\|^2$ is counted at most B times in the double summation, which lets us write

$$\frac{1}{2} \sum_{k=0}^{K-1} \sum_{k'=k-B}^{k-1} \|s(k')\|^2 \leq \frac{B}{2} \sum_{k=0}^{K-1} \|s_{p_k}\|^2. \quad (81)$$

Substituting (80) and (81) into (78) gives

$$\mathcal{F}(\mathcal{X}(0)) - \mathcal{F}(\mathcal{X}(K)) \geq \sum_{k=0}^{K-1} \sigma_k \|s_{p_k}\|^2,$$

where $\sigma_k = \gamma_k \lambda_k - \gamma_k^2 \left(\frac{L_g}{2} + \frac{\Delta L_E \delta_{\max} B}{2} \right)$. For descent to be guaranteed, we require $\sigma_k > 0$, which holds if

$$\gamma_k \lambda_k > \gamma_k^2 \left(\frac{L_g}{2} + \frac{\Delta L_E \delta_{\max} B}{2} \right).$$

That inequality is satisfied if

$$0 < \gamma_k < \frac{2\lambda_k}{L_g + \Delta L_E \delta B},$$

which is (66). ■

G. Proof of Theorem 2

Since $\mathcal{F}$ is non-negative, we can take the expectation of the cumulative descent inequality in (67) to find

$$\mathcal{F}(\mathcal{X}(0)) \geq \mathbb{E} [\mathcal{F}(\mathcal{X}(0)) - \mathcal{F}(\mathcal{X}(K))] \geq \sum_{k=0}^{K-1} \sigma_k \mathbb{E} [\|s_{p_k}\|^2],$$

and in particular

$$\mathcal{F}(\mathcal{X}(0)) \geq \sigma_{\min} \sum_{k=0}^{K-1} \mathbb{E} [\|s_{p_k}\|^2]. \quad (82)$$

We now bound the sum of expected squared gradients $\sum_{k=0}^{K-1} \mathbb{E} [\|g_k\|^2]$ in terms of this sum of squared steps $\sum_{k=0}^{K-1} \mathbb{E} [\|s_{p_k}\|^2]$. From Lemma 15, the sequence of iterates $\{\mathcal{X}(k)\}_{k \in \{0, \dots, K-1\}}$ produces non-increasing values of $\mathcal{F}$, which implies that all iterates remain within the sublevel set $\mathcal{L}(\mathcal{X}(0))$. By Lemma 1, this set is compact. Using the triangle inequality and Lemma 13, for any $k \in \{0, \dots, K-1\}$, we have

$$\begin{aligned} \|\bar{g}_k\| &\leq \|g_k\| + \|\bar{g}_k - g_k\| \\ &\leq \|g_k\| + \gamma_k \frac{\Delta L_E \delta_{\max}}{2} \sum_{k'=k-B}^{k-1} \|s(k')\|, \end{aligned} \quad (83)$$

where $\bar{g}_k = \text{grad}_{X_{p_k}} f_{p_k}(X_{p_k}(k), X_{S(p_k)}(k))$ is the local gradient that would be computed at iteration k with all robots' up-to-date pose estimates, and $g_k =$

$\text{grad}_{X_{p_k}} f_{p_k}(X_{p_k}(k), \hat{X}_{S(p_k)}(k))$ is the local gradient that robot p_k computes at iteration k with potentially outdated information. Squaring both sides of (83) and using $(a+b)^2 \leq 2a^2 + 2b^2$ gives

$$\|\bar{g}_k\|^2 \leq 2\|g_k\|^2 + 2\gamma_k^2 \left(\frac{\Delta L_E \delta_{\max}}{2} \sum_{k'=k-B}^{k-1} \|s(k')\| \right)^2. \quad (84)$$

We analyze the two terms on the right-hand side of (84) separately.

From the relation $(H_{p_k} + \lambda_k I)s_{p_k} = g_k$ in (20) and Lemma 12, we have $\|g_k\| \leq \zeta \|s_{p_k}\|$ and thus

$$2\|g_k\|^2 \leq 2\zeta^2 \|s_{p_k}\|^2. \quad (85)$$

Using the Cauchy-Schwarz inequality on the summation in (84), namely $(\sum_{j=1}^B a_j)^2 \leq B \sum_{j=1}^B a_j^2$, we find

$$\begin{aligned} 2\gamma_k^2 \left(\frac{\Delta L_E \delta_{\max}}{2} \sum_{k'=k-B}^{k-1} \|s(k')\| \right)^2 \\ \leq 2\gamma_k^2 \left(\frac{\Delta L_E \delta_{\max}}{2} \right)^2 B \sum_{k'=k-B}^{k-1} \|s(k')\|^2. \end{aligned} \quad (86)$$

Now we sum (84) from $k = 0$ to $K - 1$, take expectations, and apply (85) and (86) to find

$$\begin{aligned} \sum_{k=0}^{K-1} \mathbb{E}[\|\bar{g}_k\|^2] &\leq \sum_{k=0}^{K-1} \left(2\zeta^2 \mathbb{E}[\|s_{p_k}\|^2] \right. \\ &\quad \left. + \frac{(\Delta L_E \delta_{\max} \gamma_k)^2 B}{2} \sum_{k'=k-B}^{k-1} \mathbb{E}[\|s(k')\|^2] \right). \end{aligned} \quad (87)$$

For each k , the value of $\|s(k')\|^2$ appears in at most B of the inner summations. Therefore,

$$\sum_{k=0}^{K-1} \sum_{k'=k-B}^{k-1} \mathbb{E}[\|s(k')\|^2] \leq B \sum_{k=0}^{K-1} \mathbb{E}[\|s(k)\|^2].$$

Substituting this bound into (87) gives

$$\begin{aligned} &\sum_{k=0}^{K-1} \mathbb{E}[\|\bar{g}_k\|^2] \\ &\leq \sum_{k=0}^{K-1} 2\zeta^2 \mathbb{E}[\|s_{p_k}\|^2] + \frac{1}{2} \gamma_k^2 (\Delta L_E \delta_{\max})^2 B^2 \sum_{k=0}^{K-1} \mathbb{E}[\|s(k)\|^2] \\ &= \left(2\zeta^2 + \frac{1}{2} \gamma_k^2 (\Delta L_E \delta_{\max} B)^2 \right) \sum_{k=0}^{K-1} \mathbb{E}[\|s_{p_k}\|^2] \\ &\leq \left(2\zeta^2 + \frac{1}{2} (\Delta L_E \delta_{\max} B)^2 \right) \sum_{k=0}^{K-1} \mathbb{E}[\|s_{p_k}\|^2], \end{aligned}$$

where the equality follows from $(s(k))_q = 0$ for $q \neq p_k$ and the last line follows because $\gamma_k \leq 1$.

Let $\Upsilon \triangleq 2\zeta^2 + \frac{1}{2} (\Delta L_E \delta_{\max} B)^2$. Then

$$\sum_{k=0}^{K-1} \mathbb{E}[\|\bar{g}_k\|^2] \leq \Upsilon \sum_{k=0}^{K-1} \mathbb{E}[\|s_{p_k}\|^2]. \quad (88)$$

Rearranging (82) gives $\sum_{k=0}^{K-1} \mathbb{E}[\|s_{p_k}\|^2] \leq \frac{\mathcal{F}(\mathcal{X}^{(0)})}{\sigma_{\min}}$. Substituting this bound into (88) gives

$$\sum_{k=0}^{K-1} \mathbb{E}[\|\bar{g}_k\|^2] \leq \frac{\Upsilon \mathcal{F}(\mathcal{X}^{(0)})}{\sigma_{\min}}.$$

Since p_k is chosen uniformly at random in Algorithm 1, we have $\mathbb{E}_{p_k}[\|\bar{g}_k\|^2] = \frac{1}{R} \|\text{grad } \mathcal{F}(\mathcal{X}(k))\|^2$. Then

$$\frac{1}{R} \sum_{k=0}^{K-1} \mathbb{E}[\|\text{grad } \mathcal{F}(\mathcal{X}(k))\|^2] \leq \frac{\Upsilon \mathcal{F}(\mathcal{X}^{(0)})}{\sigma_{\min}}.$$

Replacing the sum with K times the minimum gives

$$\frac{K}{R} \min_{k \in \{0, \dots, K-1\}} \mathbb{E}[\|\text{grad } \mathcal{F}(\mathcal{X}(k))\|^2] \leq \frac{\Upsilon \mathcal{F}(\mathcal{X}^{(0)})}{\sigma_{\min}},$$

and rearranging gives the final bound of

$$\min_{k \in \{0, \dots, K-1\}} \mathbb{E}[\|\text{grad } \mathcal{F}(\mathcal{X}(k))\|^2] \leq \frac{\Upsilon R \mathcal{F}(\mathcal{X}^{(0)})}{K \sigma_{\min}}.$$